The unrestricted optimum after Conjecture 20: the one-dimensional mechanism, a width theorem, and what can and cannot be proved

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Abstract

We tried to prove that the unrestricted maximum margin equals the paper's prediction, either the interference-free (IF) value $\nu^{1-s}B_p\Gamma_s(p)$ or the price–programme value $\pi_{n,b}(p)B_p\Gamma_s(p)$ of the companion note. The outcome is as follows. (1) A universal proof of the IF value is impossible. At $p = 3$, $b = 2$ we identify the mechanism behind the counterexamples exactly: a *shared-sign, one-dimensional* family whose feature field is a rotating phase times a real exponential sum. Searching this family (exact LP for the weights, outer search over m real slopes) gives new certified networks with $\gamma/\gamma_{\mathrm{IF}} = 1.063, 1.193, 1.184, 1.457, 1.617, 1.595, 2.011, 2.212$ for $n = 11, \ldots, 18$. The excess grows with n. (2) We prove a *width theorem*: a slot-symmetric single-feature network whose input vectors share a sign pattern needs at least $\nu - \lfloor \nu/p \rfloor$ neurons. Every network in (1) attains this bound. The saving that drives the counterexamples is therefore at most a $1/p$ fraction of the neurons, and is zero when $p > \nu$, which covers the paper's experimental range. (3) For $p = 5, 7$ the same mechanism, run at its minimal width, loses to IF by a factor that grows with n (efficiency ≤ 0.19 for $p = 5$, $n \leq 16$, and decreasing). Releasing it into the full slot-symmetric class did not help at the point we tested ($n = 10$). (4) Decoupling, the remaining step of the price–programme conjecture, is *not* automatic: at $p = 5$ a mean-zero frequency-2 field carried by the same neurons raises a feature's margin by factors 1.2–2.6. In every case we found it still stays $\geq 4\times$ below IF. (5) The feature count (Section 8). Every global maximiser that buys no feature below its interference-free price uses exactly the programme's number of features, for all n, p, b. At $n = 2$ no single feature can be bought below that price at any prime, for arbitrary inputs: averaging a Riesz test over both cone edges leaves a factor $\cos(\pi/p)$ of every sign-flipped coefficient. In multi-feature maximisers, cheap features force exact arithmetic coincidences: additive relations among doubled frequencies, or modes landing on sign-flipped characters. A product of quadratic tests, one per cheap feature, detects each feature's whole L^2 interference without the $\sqrt{|L'|}$ loss of Cauchy–Schwarz. It gives a proved surplus of 39% at $n = 2$ (class $\mathcal{C}$, order-three dissociation) and reduces every n to a single-feature atom constant. The single-feature payments at $n = 2$ are proved sharp and p-free (P_{12}, and $c_1(2, s) = 1 - 3^{1-s}$, the latter computer-assisted). With them the relations case at $n = 2$ is closed for every $p \leq 59$. Near the interference-free networks the discount from interference is quadratic and the slack it forces is linear. So at $n = 2$, in the relations case and in class $\mathcal{C}$, at every prime and for any number of cheap features, no network close to an interference-free one beats the prediction (Theorem 36). In the relations case what remains is wholesale rebuilds with macroscopic, cancelling interference. Section 9 shows how the open steps fit together. The remaining steps are stated as open problems (O1)–(O4): a coverage-to-Sidon inequality, compensation costs, single-feature constants for $n \geq 3$, and $p = 5$ at large n. Every numerical test at $p \geq 5$ found no network beating the interference-free value. So the correct universal statement is the price–programme form, and the open content is concentrated in precise questions for $p \geq 5$.

1 Setting and notation

As in the paper and the companion note: p odd prime, $\rho = e^{2\pi i/p}$, $\theta := \pi/p$, $\nu = n+1$, $s = \nu/b$. A *single-feature* network has output vectors at frequency 1 only; its margin is $h = \kappa_{\min}(p)\min_x \mathrm{cm}(\Phi(x))$, where $\Phi(x) = \sum_i W_i \prod_t u_{it}(x_t)\rho^{x_t}$ and $\mathrm{cm}(z) = \mathrm{Re}\, z - \cot\theta\,|\mathrm{Im}\, z|$ (companion note, Lemmas 1–2). So $h > 0$ iff every value $\Phi(x)$ lies in the open sector $|\arg z| < \theta$. A network is *slot-symmetric* if $u_{it} = u_i$ for all t; then Φ depends on x only through the count vector $N \in \mathbb{Z}^p_{\geq 0}$, $N_a = \#\{t : x_t = a\}$:

$$\Phi(N) = \sum_{i=1}^{m} W_i \prod_{a\in\mathbb{Z}_p} \big(u_i(a)\rho^a\big)^{N_a}.$$

Efficiency means $\gamma/\gamma_{\mathrm{IF}}$ at $b = 2$ (for single-feature networks at $p \geq 5$, $n \geq 3$ this is the same as the ratio of the single-feature margin to the IF single-feature margin, because $K^*(s,p) = 1$).

2 Shared-sign networks and their fields

Definition 1. A slot-symmetric network has a *shared sign pattern* $\varepsilon \in \{\pm 1\}^p$ if $\varepsilon_a u_i(a) \geq 0$ for all i, a. The *sine pattern* is $\varepsilon_a = +1$ for $0 \leq a < p/2$ and -1 for $p/2 < a < p$, with exponents $e_a := 2a$ resp. $2a - p$, a permutation of $\{0,\dots,p-1\}$ satisfying $\varepsilon_a \rho^a = e^{i\theta e_a}$. The *one-dimensional (1-D) family* consists of the sine-pattern networks with $u_i(a) = \varepsilon_a e^{\lambda_i e_a}$ (up to scale), $\lambda_i \in \mathbb{R}$.

Lemma 2 (fields)**.** *For a sine-pattern network, $\Phi(N) = e^{i\theta\tau(N)}S(N)$ with $\tau(N) = \sum_a e_a N_a \in \{0,\dots,(p-1)n\}$ and $S(N) = \sum_i W_i \prod_a |u_i(a)|^{N_a}$, an exponential sum with positive bases. In the 1-D family*

$$\Phi(N) = e^{i\theta\tau}\sum_{i=1}^{m} V_i e^{\lambda_i \tau}, \qquad \tau = \tau(N),$$

so the margin condition is a condition on the $L+1$ values $\tau = 0,\dots,L$, $L = (p-1)n$, of a rotating phase times a real-exponential sum.

Proof. $u_i(a)\rho^a = |u_i(a)|\varepsilon_a\rho^a = |u_i(a)|e^{i\theta e_a}$; take the product over a with exponents N_a. □

At $p = 3$ the certified counterexample of the earlier notes ($n = 11$, width 8) is, to three digits, a 1-D network: all eight input vectors have the sign pattern $(-,+,+)$ (a translate of the sine pattern), and $\log|u_i(a)| - \log|u_i(0)| = \lambda_i \cdot (0,-1,1)$ with ratio -1.00 ± 0.03. Re-optimising it inside the 1-D family raises its efficiency from 1.0544 to 1.0631.

3 A width theorem

Theorem 3 (width)**.** *Let p be an odd prime, $n \geq 1$, and let a slot-symmetric single-feature network with a shared sign pattern have positive margin. Then its width is*

$$m \;\geq\; \nu - \left\lfloor \frac{\nu}{p} \right\rfloor.$$

Proof. Positive margin is an open condition, so we may perturb zero entries of the u_i to small values of the prescribed sign. Hence assume all $u_i(a) \neq 0$. Put $d = (p-1)/2$. The map $a \mapsto a + d$ is a cyclic permutation of $\mathbb{Z}_p$ of odd length, so ε cannot alternate along it, and there is a symbol

b with $\varepsilon_b = \varepsilon_{b+d}$. Let $a = b + d$ and consider the $n + 1$ count vectors $N(k) = (n - k)\mathbf{e}_b + k\mathbf{e}_a$, $k = 0, \dots, n$. With $r_i = u_i(a)/u_i(b) > 0$,

$$\Phi(N(k)) = C\left(e^{i(\pi-\theta)}\right)^k S(k), \qquad S(k) = \sum_i c_i r_i^k,$$

for constants $C \in \mathbb{C} \setminus \{0\}$, $c_i \in \mathbb{C}$, because $\rho^d = e^{i(\pi-\theta)}$. Positive margin gives $\arg S(k) \in I_k := (\omega_k - \theta, \omega_k + \theta)$ with $\omega_k = -\arg C - (\pi - \theta)k$. Note $\omega_k \equiv -\arg C + \theta k \pmod{\pi}$.

For $\psi \in \mathbb{R}$ let $h_\psi(k) = \mathrm{Im}(e^{-i\psi}S(k)) = \sum_i \mathrm{Im}(e^{-i\psi}c_i)\, r_i^k$. After merging equal bases this is a real exponential sum with at most m terms. By Laguerre's rule of signs it has at most $m - 1$ real zeros unless it vanishes identically, so the sequence $h_\psi(0), \dots, h_\psi(n)$ has at most $m - 1$ sign changes. Fix a residue k_0 and take $\psi = -\arg C + \theta k_0$. The arc I_k meets the line $\psi + \pi\mathbb{Z}$ iff $\theta(k - k_0) \in (-\theta, \theta) + \pi\mathbb{Z}$, i.e. iff $k \equiv k_0 \pmod{p}$. Call these k *undetermined*. For every other k the sign of $h_\psi(k)$ is fixed by the side of the line on which I_k lies.

- If $k, k+1$ are both determined, $I_{k+1} = I_k - \pi + \theta$; an arc of length 2θ on one side, rotated by $-\pi + \theta$ and still avoiding the line, lies on the other side, so the signs differ.
- If k is undetermined and $0 < k < n$, then $k \pm 1$ are determined ($p \geq 3$), and with I_k centred on the line, I_{k-1} and I_{k+1} are the arcs of length 2θ immediately on either side of it, so $h_\psi(k-1)$ and $h_\psi(k+1)$ again have opposite signs.

So the determined values alternate in sign and there are $|D| - 1$ sign changes, where D is the set of determined k. $h_\psi \not\equiv 0$, since otherwise every $S(k)$ would lie on the line, contradicting $D \neq \emptyset$. Choosing k_0 in a least populated residue class of $\{0, \dots, n\}$ gives $|D| = n + 1 - \lfloor \nu/p \rfloor$, hence $m - 1 \geq n - \lfloor \nu/p \rfloor$. □

Corollary 4. *(i) For $p > \nu$ a shared-sign single-feature network needs at least ν neurons, as many as the IF block. (ii) For every p, the width saving is at most $\lfloor \nu/p \rfloor$. For networks with equal amplitudes at $b = 2$, the width factor $(\nu/m)^{s-1}$ in the cost comparison with the IF block is therefore at most $\big(\nu/(\nu - \lfloor \nu/p \rfloor)\big)^{s-1} \leq \big(p/(p-1)\big)^{s-1}$: up to $(3/2)^{s-1}$ at $p = 3$, but at most $(5/4)^{s-1}$ at $p = 5$. (iii) Every certified counterexample in Table 1 attains the bound with equality.*

Remark 5. The paper's experiments use $\nu \leq 5$ and $p \geq 11$, so by (i) the mechanism behind the counterexamples has no width advantage anywhere in that range. Width is not the only lever: per unit cost, a flat sign vector has $2^{n/2}$ times the raw output of a cosine input. But numerically this lever alone never sufficed (Section 5).

4 The $p = 3$ optimum: reduction and certified values

Reduction. In the 1-D family, for fixed slopes $\lambda_1, \dots, \lambda_m$ the cone constraints at $\tau = 0, \dots, L$ are linear in $(\mathrm{Re}\, V_i, \mathrm{Im}\, V_i)$, and the cost amplitude of neuron i is $\sqrt{p/2}\,|V_i|\, Z(\lambda_i)^n$ with $Z(\lambda)^2 = \sum_{k<p} e^{2\lambda k}$. Maximising the margin under a weighted ℓ_1 budget is therefore a second-order cone programme (an LP after polygonal approximation of $|V_i|$). The concave $\ell_{2/\nu}$ cost is handled by reweighting; in practice the optimal amplitudes are nearly equal and uniform weights suffice. The whole problem reduces to choosing m real slopes. We search them with CMA-ES and continuation in n. All values below are then re-verified independently.

Structure of the optimum. Each term sits at the cost-optimal tilt for the stretch of τ it covers. The cost amplitudes agree to within $\sim 10\%$, consecutive terms have nearly antipodal phases ($\approx 172^\circ$ apart at $n = 11$), and the margin is active at most τ. The field is a 10^{-3}–10^{-4} residue of terms up to 10^3 times larger, kept inside the 60° sector at every count class.

n	11	12	13	14	15	16	17	18
width m (bound $\nu - \lfloor \nu/3 \rfloor$)	8 (8)	9 (9)	10 (10)	10 (10)	11 (11)	12 (12)	12 (12)	13 (13)
$\gamma/\gamma_{\mathrm{IF}}$	1.06308	1.19335	1.18360	1.45687	1.61704	1.59546	2.01105	2.21197
previously certified	1.0544	–	1.1091	1.4293	–	–	–	–

Table 1: $p = 3$, $b = 2$. All values are certified by `certify_1d.py`: weights are rounded to ten significant digits, the margin is computed in exact rational arithmetic from the logits over all count vectors (hence over all 3^n inputs), and $\gamma/\gamma_{\mathrm{IF}}$ is enclosed by interval arithmetic.

Hence $\pi_{n,2}(3)\,\nu^{s-1}$ is at least 1.06, 1.19, 1.18, 1.46, 1.62, 1.60, 2.01, 2.21 for $n = 11, \ldots, 18$. The interference-free prediction underestimates the $p = 3$ optimum by a factor that, numerically, grows by about 10% per unit of n.

5 $p \geq 5$: the mechanism loses

Gain/loss accounting. For a 1-D network with nearly equal amplitudes, $\gamma/\gamma_{\mathrm{IF}} = \zeta \cdot 2^{n/2} \cdot (\nu/m)^{s-1}$. Here $2^{n/2}$ is the raw advantage of flat sign vectors over cosines, $(\nu/m)^{s-1}$ the width saving (capped by Theorem 3), and ζ the fraction of the mid-range envelope that survives cancellation. The available gain per unit of n is $\frac{1}{2}\log\frac{2p}{p-1}$: 0.55 at $p = 3$, 0.46 at $p = 5$, 0.42 at $p = 7$. Numerically, $\log\zeta/n \approx -0.48$ at $p = 3$ ($n = 18$) and ≈ -0.575 at $p = 5$ (stable over $n = 13$–17). That leaves net rates of about $+0.07$ and -0.12 per unit of n, consistent with the observed growth at $p = 3$ and decline at $p = 5$.

n	6	7	8	9	10	11	12	13	14	15	16
$p = 5$	.192	.179	.166	.147	.132	.127	.126	.120	.108	.097	.093
width (bound)	6 (6)	7 (7)	8 (8)	9 (8)	10 (9)	10 (10)	11 (11)	12 (12)	13 (12)	14 (13)	14 (14)
$p = 7$	.126	.107	.089	.076	.066	.063	.058	.049	.044	.035	–

Table 2: Best 1-D efficiency $\gamma/\gamma_{\mathrm{IF}}$ found (slope search with exact LP, continuation in n with insertion and deletion moves and fresh global candidates at every n).

The 1-D restriction is not the bottleneck. At $p = 5$, $n = 10$, released into the full slot-symmetric single-feature class (arbitrary input vectors, `cone.py`), the 1-D optimum 0.1318 is a local optimum (trial with no noise: 0.1318; with noise: lower). Mixing in positive-input neurons, which keeps the field a function of τ but breaks the shared sign pattern, lowered the efficiency at $p = 3$, $n = 11$ ($1.063 \to 0.82$).

6 Cross-feature compensation: decoupling is not automatic

At $p \geq 5$ several frequencies coexist, and the margin condition at x is that the vector $(\Phi_l(x))_l$ lies in a polyhedron, not that each $\Phi_l(x)$ lies in its own sector. Concretely, at $p = 5$ a frequency-2 value $\Phi_2 \approx 0.25|\Phi_1|e^{-i18^\circ}$ makes Φ_1 admissible at $\arg\Phi_1 = 60^\circ$, the $p = 3$ sector. The same neurons can

produce such a Φ_2 almost for free: the field is a 10^{-3} residue of much larger terms, so a frequency-2 output component with $\sim 10^{-3}$ relative weight, costing $\sim 10^{-6}$ of the budget, gives a Φ_2 of the right size. It is limited only by geometry: its phase follows a different linear functional of the counts.

We tested this with an exact LP over output vectors spanning frequencies 1 and 2, at the slopes of the best 1-D networks, on all count classes and all four shifts δ. Measured gains over frequency-1 outputs at the same neurons: $\times 1.17$ ($n = 8$), $\times 1.36$ ($n = 12$), up to $\times 2.6$ at non-optimal widths. With a CMA search over slopes for the joint problem: 0.197 ($n = 8$) and 0.173 ($n = 12$). The frequency-2 output carries 0.2–3% of the output energy. So the decoupling half of the price–programme conjecture (companion note, Conjecture 20′) cannot be proved feature by feature: one feature can live outside its own sector where another, interference-only, field compensates. In every configuration we tried, the compensated feature still stays $\geq 4\times$ below IF.

7 The number of features of the unrestricted optimum

The paper's target is the *count*: a frequency j is a feature of θ if $A_j(\theta) = |\beta_j(\theta)| > 0$, and the prediction is that every globally margin-optimal network uses $K^*(s,p) = |\operatorname{spt} y^\star|$ features, with amplitudes proportional to a maximiser $y^\star$ of the programme $\Gamma_s(p)$. This is weaker than the value prediction, and it survives the counterexamples.

Proposition 6 (what is unconditional)**.** *Let $\theta^\star$ be a global maximiser of the $L_{2,b}$-margin.*

(a) If $p = 3$, then $\theta^\star$ uses exactly $K^*(s,3) = 1$ feature, for every n and b.

(b) If $\theta^\star$ is interference-free, its amplitudes are proportional to a maximiser of $\Gamma_s(p)$, so it uses exactly $|\operatorname{spt} y^\star|$ features (paper, Theorem 8(1)).

(c) If $b_0 \leq b \leq \nu$ (interval of the earlier notes, Part II), then $\theta^\star$ is interference-free and uses all $F = K^*(s,p)$ frequencies with equal amplitudes.

(d) For every p, n, b: write $\eta(\theta) := \|A(\theta)\|_{1/s}/(B_p\|\theta\|_{2,b}^\nu) \leq 1$ for the clean efficiency and $y = A/\|A\|_{1/s}$ for the clean allocation of $\theta^\star$. Then $\eta(\theta^\star) \geq \pi_{n,b}(p)$ and $\min_{\delta\neq 0}\sum_j y_j\kappa_j(\delta) \geq \frac{\pi_{n,b}(p)}{\eta(\theta^\star)}\,\Gamma_s(p)$. In particular, if $\eta(\theta^\star) = \pi_{n,b}(p)$, then y is a programme maximiser and $\theta^\star$ uses exactly $|\operatorname{spt} y|$ features.

Proof. (a) $F = \{1\}$, and $A_1 > 0$ by the paper's Lemma 2 because the margin is positive. (b), (c) are the cited results. (d) By the price–programme theorem of the companion note, $h(\theta^\star) = \gamma_b^*\|\theta^\star\|^\nu \geq \pi B_p\Gamma_s\|\theta^\star\|^\nu$. By the paper's Lemma 2, $h \leq \min_\delta \sum_j A_j\kappa_j(\delta) = \|A\|_{1/s}\min_\delta\sum_j y_j\kappa_j(\delta) \leq \|A\|_{1/s}\Gamma_s$. Divide by $B_p\|\theta^\star\|^\nu$. □

Proposition 7 (decoupling suffices; the price is irrelevant)**.** *Suppose a global maximiser $\theta^\star$ satisfies the decoupling inequality*

$$h(\theta^\star) \;\leq\; \pi_{n,b}(p)\,B_p\,\min_{\delta\neq 0}\sum_j P_j^s\kappa_j(\delta) \qquad \text{for some } P_j \geq 0,\;\; \sum_j P_j \leq \|\theta^\star\|_{2,b}^b.$$

Then $(P_j^s/\|\theta^\star\|_{2,b}^\nu)_j$ is a maximiser of $\Gamma_s(p)$. If moreover the features of $\theta^\star$ are exactly the frequencies it pays for ($A_j > 0 \iff P_j > 0$), then $\theta^\star$ uses exactly K^* features.

Proof. With $z_j = P_j^s/\|\theta^\star\|^\nu$ we have $\sum_j z_j^{1/s} \leq 1$, and

$$\pi B_p\Gamma_s\|\theta^\star\|^\nu \leq h(\theta^\star) \leq \pi B_p\|\theta^\star\|^\nu \min_\delta \sum_j z_j\kappa_j(\delta).$$

Hence z attains Γ_s. □

The price enters only as a common factor, because relabelling $a \mapsto ja$ makes it the same for every frequency and scaling makes it homogeneous of degree s in the budget. So the count needs *only* decoupling. Neither $\pi = \nu^{1-s}$ nor interference-freeness is needed.

Theorem 8 (exact count in the dense regime; lower bound in general)**.** *For $S \subseteq F$ let $\Gamma_s^{(S)}(p)$ be the programme restricted to allocations supported on S, and $K_{\mathrm{low}}(s,p,\nu) := \min\{|S| : \Gamma_s^{(S)}(p) \geq \nu^{1-s}\Gamma_s(p)\}$. Every global maximiser of the $L_{2,b}$-margin, of any width and with interference allowed, uses at least K_{low} features. In particular, if*

$$\nu^{s-1}\,\Gamma_s^{(F\setminus\{j\})}(p) < \Gamma_s(p) \qquad (\textit{independent of } j \textit{ by relabelling}),$$

every global maximiser uses all $F = (p-1)/2$ frequencies. This is the exact count $K^*(s,p) = F$ in the dense regime. The condition holds for $s \leq 1.331, 1.285, 1.257$ ($p = 5$; $\nu = 3, 4, 5$), $s \leq 1.154, 1.130, 1.115$ ($p = 7$), $s \leq 1.079, 1.066, 1.058$ ($p = 11$), $s \leq 1.064, 1.053, 1.047$ ($p = 13$). In every case computed this is 2.5 to 5 times the range $s - 1 \leq 1/(F(2\nu+1) - 2)$ of Part II.

Proof. If the features of $\theta^\star$ lie in S, Proposition 3 of the paper restricted to S and the single-neuron budget give $\gamma(\theta^\star) \leq B_p\Gamma_s^{(S)}$, while $\gamma(\theta^\star) = \gamma_b^* \geq \nu^{1-s}B_p\Gamma_s$ (Theorem 8). Relabelling $a \mapsto ka$ maps allocations on S to allocations on kS with the same value, and acts transitively on F. □

Why the upper bound is the hard half. Let q be KKT multipliers of a maximiser on its active constraints (x,δ). Stationarity in the output vector makes neuron i's frequency-j output proportional to $\sum_x g_i(x)\rho^{-jy(x)}\overline{Q_j(x)}$, with $g_i(x) = \prod_t u_{it}(x_t)$ and $Q_j(x) = \sum_\delta q(x,\delta)(1-\rho^{-j\delta})$. If q is uniform in x, as for any interference-free maximiser (average q over translations), this gives $\beta_{ij} \propto Q_j\,|\sum_x g_i(x)\rho^{-jy(x)}|^2$ with $\operatorname{Re} Q_j = \sum_\delta q(\delta)\kappa_j(\delta) > 0$. Then every frequency at which some neuron's input product has a non-zero clean projection is a feature. So "no more than K^* features" is equivalent to frequency purity of the maximiser's input products at the non-programme frequencies. Interference-free maximisers have it (paper, Theorem 8(2)). A maximiser built from sign-vector inputs, like the $p = 3$ networks, would generically carry every frequency. The upper bound is therefore not separable from the question of which input structure is optimal at $p \geq 5$.

Theorem 9 (the count is $\Theta(\log p)$ for every near-optimal network)**.** *Let $2 \leq b < \nu$, $s = \nu/b$, $r = 1/s$, and for a non-zero amplitude vector A let*

$$K_r(A) := \frac{\big(\sum_j A_j^r\big)^{1/(1-r)}}{\big(\sum_j A_j\big)^{r/(1-r)}} \;\leq\; |\operatorname{spt} A|,$$

the participation number of A (it equals K for K equal non-zero entries). Let θ be any product network, of any width and with interference allowed, with $\gamma(\theta) \geq \lambda\,\nu^{1-s}B_p\Gamma_s(p)$ for some $\lambda \in (0,1]$. Every global maximiser has $\lambda \geq \pi_{n,b}(p)\nu^{s-1} \geq 1$. Then

$$K_r(A(\theta)) \;\leq\; \nu\Big(\frac{p}{(p-1)\lambda\,\Gamma_s(p)}\Big)^{\frac{1}{s-1}} \;\leq\; \nu\Big(\frac{2p}{(p-1)\lambda}\Big)^{\frac{1}{s-1}}\lceil 8\ln p\rceil.$$

Moreover, for every $M \geq 1$ the M largest amplitudes carry at least a fraction $1 - (K_r/M)^{1-r}$ of $\sum_j A_j$. Together with the paper's Corollary 12, which shows that the support of such a network has at least $c_{\nu,b,\lambda}\ln p$ frequencies, the feature count of every margin-optimal network is $\Theta_{\nu,b}(\log p)$. This is the order of $K^*(s,p)$ (paper, Theorem 11), for every odd prime, unconditionally.

Proof. Averaging the paper's Lemma 2 over $\delta \neq 0$ and using $\sum_{\delta\neq 0} \kappa_j(\delta) = p$ gives $h \le \frac{p}{p-1}\sum_j A_j$. The single-neuron budget gives $\sum_j A_j^r \le \sum_{i,j} |\beta_{ij}|^r \le C_b P$ with $P = \|\theta\|_{2,b}^b$. Hence, with $h = \gamma P^s$ and $sr = 1$,

$$K_r(A) \le \frac{(C_b P)^{1/(1-r)}}{\left(\frac{p-1}{p}\gamma P^s\right)^{r/(1-r)}} = \left(\frac{p\, C_b^{1/r}}{(p-1)\gamma}\right)^{\frac{r}{1-r}} = \left(\frac{p\, B_p}{(p-1)\gamma}\right)^{\frac{1}{s-1}},$$

because $C_b^{1/r} = C_b^{\nu/b} = B_p$ and $r/(1-r) = 1/(s-1)$. Insert $\gamma \ge \lambda \nu^{1-s} B_p \Gamma_s$, then $\Gamma_s \ge \frac{1}{2}\lceil 8 \ln p\rceil^{1-s}$ (paper, Theorem 10). The bound $\lambda \ge \pi \nu^{s-1}$ for maximisers is the price–programme theorem. For the tail statement, normalise $\sum_j A_j = 1$ and sort; the amplitudes beyond the M-th are at most $1/M$ each, so their mass t satisfies $tM^{1-r} \le \sum_{j>M} A_j^r \le K_r^{1-r}$. □

Corollary 10 ($n = 2$, every prime $p \ge 5$)**.** *For $n = 2$ and $2 \le b \le 3$, no single-frequency network (inputs in $\operatorname{span}\{\mathbf{1}, \cos, \sin\}$ of one frequency, any width) beats the interference-free single-feature value at any prime $p \ge 5$.*

Proof. For $n = 2$ the robust constant of the companion note's Theorem 14 is $k^*(2, s) = \sup(\operatorname{Re} A - 3^{1-s}(\sum |\beta_i|^{1/s})^s)/(R/\sqrt{2})$ with $R^2 = |B|^2 + |C|^2 + |D|^2$, where B, C, D are the three sign-flipped sums. The four-moment inequality of the earlier notes (Part I, Theorem A: $|A| - R/(2\sqrt{3}) \le 3^{1-s}(\sum |\beta_i|^{1/s})^s$ for $1 \le s \le 3/2$, valid for arbitrary unit phases) gives $k^*(2, s) \le \sqrt{2}/(2\sqrt{3}) = 1/\sqrt{6} < 0.41 < k_5 = 0.946 \le k_p$. The companion note's Theorem 14 then applies. □

Lemma 11 (slack identity)**.** *For every logit table (any network, any p, n, any inputs), with margin h and slack $\sigma(x, \delta) := f(x)[y(x)] - f(x)[y(x) - \delta] - h \ge 0$,*

$$\Phi_j(x) = \tfrac{2}{p}\Big(h - \sum_{\delta\neq 0} \sigma(x, \delta)\rho^{j\delta}\Big) \qquad (j \in F).$$

Consequently every sign-flipped coefficient satisfies $|\beta_{j,T}| \le \frac{2}{p}S$ with $S := \mathbb{E}_x \sum_\delta \sigma = p\sum_{j\in F} \operatorname{Re}\beta_j - (p-1)h$. The same slack S bounds the interference of every *frequency.*

Proof. $\hat{f}_x(j) = \sum_\delta f(x)[y - \delta]\rho^{-j(y-\delta)} = -\rho^{-jy}\sum_{\delta\neq 0}(h + \sigma(x, \delta))\rho^{j\delta}$ because $\sum_\delta \rho^{j\delta} = 0$, and $-\sum_{\delta\neq 0}\rho^{j\delta} = 1$. Then $\beta_{j,T} = \overline{\mathbb{E}_x \Phi_j \chi}$ for a unimodular character χ, and $|\mathbb{E}[\sigma\chi]| \le \mathbb{E}\sigma$. For S, use $\mathbb{E}_x \Phi_j = \overline{\beta_j}$ and that $\sum_{j\in\mathbb{Z}_p} \hat{\bar{\sigma}}(j) = p\,\bar{\sigma}(0) = 0$. □

Proposition 12 ($n = 2$, any inputs)**.** *For $n = 2$, $2 \le b \le 3$, every prime p and every network, $h \le \frac{p}{p-1}\big(\sum_j \operatorname{Re}\beta_j - \max_j R_j/(2\sqrt{3})\big)$, where $R_j^2 = \sum_{T\neq\emptyset} |\beta_{j,T}|^2$. For single-feature networks (any inputs) this gives $\gamma \le \frac{p}{p-1}\nu^{1-s}B_p$, i.e. $\pi_{2,b}(p) \le \nu^{1-s} \cdot \frac{p}{(p-1)\kappa_{\min}(p)}$. This is Part I's sharp result at $p = 3$, but it is weaker than the IF value at $p \ge 5$.*

Proof. Lemma 11 with three non-empty T gives $R_j \le 2\sqrt{3}(\sum_{j'} \operatorname{Re}\beta_{j'} - \frac{p-1}{p}h)$. For one feature, apply Part I's Theorem A ($|A| - R/(2\sqrt{3}) \le 3^{1-s}(\sum |\beta_i|^{1/s})^s$; Lemma 6 of the paper supplies the unit phases for arbitrary inputs) and the single-neuron budget. □

Two losses separate Proposition 12 from the exact answer at $p \ge 5$, and both are the multi-frequency obstacle. (i) A single feature's gaps at its non-binding offsets carry structural slack, which uses up S without any interference. (ii) With several frequencies, one slack budget bounds each R_j but not $\sum_j R_j$, so only one frequency's interference is charged. Part II of the earlier notes overcomes (ii) only near $b = \nu$, by mixing the chain matrices of all frequencies.

The general $n = 2$ case (inputs with other frequencies) is also where the L^2 route stops. Input energy ε outside ± 1 costs only $O(\varepsilon)$ of clean amplitude, but it can create non-circular (partner-mode) interference of size $O(\sqrt{\varepsilon})$. So every L^2/moment bound fails in a thin regime next to the interference-free point, and a pointwise argument, like Part I's at $p = 3$, is required.

Numerical tests of the count with general networks. (i) *Width.* For slot-symmetric networks with arbitrary input vectors (any sign patterns) and outputs on all frequencies, we searched for the smallest width with positive margin (elimination from feasible starts, exact LP for the outputs). It equals $\nu - \lfloor \nu/p \rfloor$ in every case tested: $p = 3$, $n = 3, 4, 5, 6, 8$ (widths $3, 4, 4, 5, 6$), and $p = 5$, $n = 4, 5, 6, 7$ (widths $4, 5, 6, 7$), with one output frequency or all of them. So mixing frequencies does not save neurons, and Theorem 3 appears to hold for all slot-symmetric networks. (ii) *Allocation.* At $(p, n, b) = (7, 3, 2)$ the programme has $K^* = 2$ with unequal shares $(0.328, 0, 0.182)$. Global search (inputs by CMA-ES from interference-free blocks on every frequency subset plus perturbations, outputs by exact conic programme with ℓ_r reweighting) gave: one frequency 0.801; any two frequencies 1.00000 with amplitude shares equal to the programme's up to relabelling, unchanged under perturbation; all three frequencies 0.827. At $(5, 2, 2)$ ($K^* = 2$), random-start search reached 0.755 (width 4, one feature), 0.821, 0.941, 0.970 (widths 5–7, two features with near-equal shares), and never exceeded the K^*-feature block.

Where a proof stops. Combining (a)–(c): the count prediction holds for every global optimiser, *except possibly* when $p \geq 5$, $b < b_0$ and the optimiser carries interference. None of the known counterexamples is in that set, since they all have $p = 3$. Closing the remaining case requires a price-sharp upper bound, i.e. Conjecture $20'$ or an equivalent. The only unconditional upper bound, Theorem 4 of the paper, is too weak: it loses the factor $\pi\nu^{s-1}$, which swamps the differences between allocations (removing one feature changes Γ_s by a factor $(1 - w_j)^s$ only, paper Lemma 19). The compensation of Section 6 is the precise danger. It uses frequencies whose clean amplitude is tiny but non-zero and that the programme does not pay for. An optimiser built that way would *over*count, so the strict count statement at $p \geq 5$ is exactly as hard as excluding compensation at the optimum. In every experiment at $p = 5, 7$, compensated networks stay $\geq 4\times$ below the K^*-feature IF block. Section 8 proves the count for a class that carries interference, and isolates what is left.

8 A Lipschitz form of Lemma 7 and a transport reduction

Lemma 13 (robust interference-free bound, every n)**.** *Let $0 < r \leq 1$, $s = 1/r$, $\beta_1, \ldots, \beta_N \in \mathbb{C}$, $\xi_l \in \mathbb{T}^n$, $A = \sum_l \beta_l$, $\beta_T = \sum_l \beta_l \prod_{t \in T} \xi_{lt}$, $Q = \sum_l |\beta_l|^r$, and let $\ell := \max |\beta_T|$ over the non-empty* intervals $T = \{t' + 1, \ldots, t\} \subseteq [n]$. *Then*

$$\textit{(i)}\ \ |A|^r \leq \nu^{r-1} Q + \|L\|_{\mathrm{op}}^r, \qquad \textit{(ii)}\ \ |A| \leq \nu^{1-s} Q^s + K_{\nu,s}\, \ell, \quad K_{\nu,s} := \tfrac{\nu! - 1}{\nu}\, \nu^{(s-1)(\nu-1)},$$

where L is the strictly lower part of the chain matrix. With $\ell = 0$ both are the paper's Lemma 7.

Proof. Let $H = \sum_l \beta_l m_l m_l^*$ with the chain vectors m_l of the paper's Lemma (chain). Its diagonal is A, its strictly lower entries are the interval sums β_T, so $H = H_0 + L$ with $H_0 = AI + U$ upper triangular and $|L_{tt'}| \leq \ell$. Rotfel'd's inequality for $0 < r \leq 1$ gives $\sum_k \sigma_k(H)^r \leq \sum_l \|\beta_l m_l m_l^*\|_{S_r}^r = \nu^r Q$. (i) All eigenvalues of H_0 equal A, so Weyl's majorant theorem gives $\sum_k \sigma_k(H_0)^r \geq \nu |A|^r$. Weyl's perturbation bound $\sigma_k(H_0) \leq \sigma_k(H) + \|L\|$ and subadditivity of $t \mapsto t^r$ give $\nu |A|^r \leq \nu^r Q + \nu \|L\|^r$. (ii) Every non-identity permutation has a factor below the diagonal, and every entry

of H has modulus at most $S_1 := \sum_l |\beta_l| \le Q^s$. So $|\det H - A^\nu| \le (\nu! - 1)\,\ell\, S_1^{\nu-1}$. By AM–GM on σ_k^r, $|\det H| \le (\nu^{-1}\sum_k \sigma_k^r)^{\nu/r} \le a^\nu$ with $a := \nu^{1-s}Q^s$. Hence $|A|^\nu \le a^\nu + (\nu! - 1)\ell Q^{s(\nu-1)}$, and $(a + \varepsilon/(\nu a^{\nu-1}))^\nu \ge a^\nu + \varepsilon$ gives (ii), since $Q^{s(\nu-1)}/a^{\nu-1} = \nu^{(s-1)(\nu-1)}$. □

By the paper's Lemma 6 the lemma applies to the neurons of any network at any frequency j, with $A = \beta_j$ and $\ell = \ell_j := \max_{T \text{ interval}} |\beta_{j,T}|$.

Corollary 14 (single-frequency networks, every n, all large p)**.** *The constant of the companion note's Theorem 14 satisfies $k^*(n,s) \le \sqrt{2}\,K_{\nu,s} < \infty$ for every n and s. Since $k_p = \sqrt{\cot^2(\pi/p) - 1} \ge p/\pi - 1$ for $p \ge 5$, no single-frequency network beats the interference-free single-feature value at any prime $p \ge \pi(1 + \sqrt{2}K_{\nu,s})$. (For $n = 2$, Corollary 10 already gives every $p \ge 5$.)*

Proof. $\operatorname{Re} A - \nu^{1-s}Q^s \le K_{\nu,s}\ell \le K_{\nu,s}\sqrt{E} = \sqrt{2}K_{\nu,s}\sqrt{E/2}$. For k_p: $\cot x \ge 1/x - x/2$ on $(0, \pi/5]$ (the difference vanishes at 0 and has derivative $1/x^2 + 1/2 - \csc^2 x \ge 0$ there), so $\cot^2 x - 1 \ge x^{-2} - 2 \ge (x^{-1} - 1)^2$ for $x \le 2/3$. □

The constant is crude ($K_{3,3/2} = 5$, $K_{4,2} = 368$), so the threshold is large, but it is finite and explicit for every n.

Definition 15. A network is *frequency-pure without input DC* (class $\mathcal{C}$) if every neuron i has a frequency $j_i \in F$ with $u_{it} \in \operatorname{span}\{\cos(2\pi j_i \cdot /p), \sin(2\pi j_i \cdot /p)\}$ for all t and $w_i \in \operatorname{span}\{\mathbf{1}, \cos(2\pi j_i \cdot /p), \sin(2\pi j_i \cdot /p)\}$. Neurons of different frequencies may coexist in any numbers. Every interference-free block of the paper is in $\mathcal{C}$.

For $\theta \in \mathcal{C}$, expanding $u(a)\rho^{la} = p^{-1}(\hat{u}(-l) + \hat{u}(l)\rho^{2la})$ gives $\Phi_l(x) = \sum_{T \subseteq [n]} a_{l,T}\,\rho^{2l s_T(x)}$ with $s_T(x) = \sum_{t \in T} x_t$, $a_{l,\emptyset} = \overline{\beta_l}$ and $|a_{l,T}| = |\beta_{l,T}|$ (companion note, Lemma 11). For $T \ne \emptyset$ define the *interference profile*

$$\varphi_T(w) := \sum_{l \in F} \operatorname{Re}\left(a_{l,T}\rho^{lw}\right), \qquad w \in \mathbb{Z}_p,$$

a mean-zero real function on $\mathbb{Z}_p$ whose Fourier coefficients are the sign-flipped coefficients of all features at once. Write $m_\delta := \mathbb{E}_x g(x,\delta) - h \ge 0$ for the mean slack and $\bar{m}_\delta = \frac{1}{2}(m_\delta + m_{-\delta})$.

Lemma 16 (interference profiles are slack potentials)**.** *Let $\theta \in \mathcal{C}$, p any odd prime. For every non-empty T, every $w \in \mathbb{Z}_p$ and every $\delta \ne 0$,*

$$\varphi_T(w) - \varphi_T(w - \delta) \;\ge\; -m_\delta.$$

Consequently $|\beta_{l,T}|\,|1 - \rho^{l\delta}| \le 2m_\delta$ for all l, T, δ, i.e. $|\beta_{l,T}| \le \sqrt{2}\,\min(m_\delta, m_{-\delta})/\sqrt{\kappa_l(\delta)}$.

Proof. With $c_l(\delta) = 1 - \rho^{-l\delta}$, $\operatorname{Re}(a_{l,T}c_l(\delta)\rho^{2ls_T}) = \operatorname{Re}(a_{l,T}\rho^{l\cdot 2s_T}) - \operatorname{Re}(a_{l,T}\rho^{l(2s_T - \delta)})$, so by the companion note's Lemma 2

$$g(x,\delta) = \mathbb{E}_x g(\cdot,\delta) + \sum_{T \ne \emptyset} \big(\varphi_T(2s_T(x)) - \varphi_T(2s_T(x) - \delta)\big).$$

Fix $T_0 \ne \emptyset$ and condition on $s_{T_0}(x) = z$. For $T \ne T_0$ the forms s_T, s_{T_0} are linearly independent over $\mathbb{Z}_p$, so s_T is conditionally uniform and φ_T averages to 0. Hence $h \le \mathbb{E}[g \mid s_{T_0} = z] = h + m_\delta + \varphi_{T_0}(2z) - \varphi_{T_0}(2z - \delta)$; put $w = 2z$ (p odd). For the consequence, $w \mapsto m_\delta + \varphi_T(w) - \varphi_T(w - \delta)$ is non-negative with mean m_δ, and its coefficient at ρ^{lw} is $\frac{1}{2}a_{l,T}c_l(\delta)$, with $|c_l(\delta)| = \sqrt{2\kappa_l(\delta)}$. □

So the pointwise margin constraints, projected onto each T, say exactly that $\varphi_T(w') - \varphi_T(w) \le m_{w-w'}$: the interference profile is a potential for the slack, i.e. 1-Lipschitz for the path quasi-metric D_m that the mean slacks induce on the Cayley graph of $\mathbb{Z}_p$. By Kantorovich–Rubinstein duality, the largest coefficient a feature l can carry at T is a transport cost: $\max|\beta_{l,T}| = \frac{2}{p}\max_\phi W_{D_m}(\nu^+_{l,\phi}, \nu^-_{l,\phi})$, $\nu_{l,\phi}(w) = \cos(2\pi lw/p + \phi)$. No circularity is used, so this holds at $p = 3$ as well.

Theorem 17 (decoupling when few features are bought cheaply)**.** *Let $\theta \in \mathcal{C}$, let $a_l := \nu^{1-s}Q_l^s$ with $Q_l = \sum_i |\beta_{il}|^r$ be the interference-free price of the amplitudes θ pays for, let $\delta^\star$ minimise $\sum_l a_l\kappa_l(\delta)$, and let $L' = \{l : |\beta_l| > a_l\}$ be the frequencies bought* below *the interference-free price. If*

$$\sum_{l\in L'} \kappa_l(\delta^\star) \;\le\; \frac{2}{(2^n-1)\,k^*(n,s)^2} \qquad (= 4 \text{ for } n = 2),$$

then $\gamma(\theta) \le \nu^{1-s}B_p\Gamma_s(p)$, and if the inequality is strict, equality forces θ to be interference-free. For $n = 2$ the condition holds whenever $|L'| \le 2$, because $\kappa_l < 2$. Hence, for $n = 2$, every odd prime p and every $b \in [2, 3]$: among networks of $\mathcal{C}$ in which at most two frequencies are bought below the interference-free price (any width, any interference elsewhere), the maximum is the interference-free value, and every maximiser is interference-free and uses exactly $|\operatorname{spt} y^\star| = K^*(s,p)$ features.

Proof. Averaging $g(\cdot, \pm\delta)$ over x gives, for every δ, $h = \sum_l \operatorname{Re}\beta_l\,\kappa_l(\delta) - \bar m_\delta$. By the companion note's Theorem 14 (whose constant is finite by Corollary 14), $|\beta_l| - a_l \le k^*\sqrt{E_l/2}$ with $E_l = \sum_{T\neq\emptyset}|\beta_{l,T}|^2$. For $n = 2$ take $k^* = 1/\sqrt{6}$ (Part I, Theorem A). Fix T. By the proof of Lemma 16, $D_T(w) := \varphi_T(w) - \varphi_T(w - \delta^\star)$ has mean zero and values in $[-m_{\delta^\star}, m_{-\delta^\star}]$ (the upper bound is the constraint at $(w - \delta^\star, -\delta^\star)$). So $\frac{1}{p}\sum_w D_T^2 \le m_{\delta^\star}m_{-\delta^\star} \le \bar m^2$ (Bhatia–Davis), and Parseval, with coefficients $a_{l,T}c_l(\delta^\star)$ and $|c_l|^2 = 2\kappa_l$, gives $\sum_l |\beta_{l,T}|^2\kappa_l(\delta^\star) \le \bar m^2$. Summing over T: $\sum_l E_l\kappa_l(\delta^\star) \le (2^n - 1)\bar m^2$. By Cauchy–Schwarz,

$$h \le \sum_l a_l\kappa_l(\delta^\star) + \frac{k^*}{\sqrt{2}}\sum_{l\in L'}\sqrt{E_l}\,\kappa_l(\delta^\star) - \bar m \le \sum_l a_l\kappa_l(\delta^\star) + \bar m\Big(k^*\sqrt{\tfrac{2^n-1}{2}\textstyle\sum_{l\in L'}\kappa_l(\delta^\star)} - 1\Big).$$

Under the condition the bracket is ≤ 0, so $h \le \nu^{1-s}\min_\delta\sum_l Q_l^s\kappa_l(\delta) \le \nu^{1-s}(C_bP)^s\Gamma_s$ by the single-neuron budget $\sum_l Q_l \le C_bP$. If the bracket is < 0 and equality holds, then $\bar m = 0$, and Lemma 16 gives $\beta_{l,T} = 0$ for all l, T. So the maximiser is interference-free, and the paper's Theorem 8 gives its amplitudes and its count. □

Theorem 17 is the first count result at $p \ge 5$ outside the dense regime for a class of networks that carry interference. Any number of features may carry interference, and up to two of them (for $n = 2$) may use it to buy themselves below the interference-free price. They still cannot beat the interference-free value, because the slack their interference forces at $\delta^\star$ is at least as large as what they gain. What it does not cover is *collective* undercutting: many features each bought a little below price, with phases arranged so that their profiles nearly cancel. The proof charges these through Cauchy–Schwarz over L', which loses $\sqrt{|L'|}$. A proof of the general count must replace this step by an estimate that sees the cancellation.

The count fails only through cheap features. The next proposition holds for every n, p, b and every network. It says that the count can fail only if the maximiser buys some frequency *below* its interference-free price, so everything reduces to excluding cheap features.

Proposition 18 (no cheap features ⇒ exact count)**.** *Let $\theta^\star$ be a global maximiser of the $L_{2,b}$-margin, $a_l = \nu^{1-s}Q_l^s$, $Q_l = \sum_i|\beta_{il}|^r$. If $|\beta_l| \le a_l$ for every l, then $(Q_l^s)_l$ is proportional to a maximiser $y^\star$ of $\Gamma_s(p)$, $|\beta_l| = a_l$ for every l, and $\theta^\star$ uses exactly $|\operatorname{spt} y^\star|$ features.*

Proof. $h \le \min_\delta \sum_l \operatorname{Re}\beta_l\kappa_l(\delta) \le \nu^{1-s}\min_\delta \sum_l Q_l^s\kappa_l(\delta) \le \nu^{1-s}(\sum_l Q_l)^s\Gamma_s \le \nu^{1-s}(C_bP)^s\Gamma_s$, and the left side is at least the right side because interference-free networks attain it (paper, Theorem 8). So every inequality is an equality. The third makes (Q_l^s) proportional to a programme maximiser. The first two give $|\beta_l| = a_l$ wherever $\kappa_l(\delta) > 0$, i.e. everywhere, so $\beta_l \neq 0 \iff Q_l > 0$. □

Proposition 18 does not need the maximiser to be interference-free. At $p = 3$ the counterexamples buy their single feature below price, consistent with the proposition.

Theorem 19 (Riesz products: cheap features need arithmetic coincidences, $n = 2$)**.** *Let $n = 2$, $2 \le b \le 3$, p any odd prime, and θ any network (inputs and outputs arbitrary). Write $\Phi_l(x) = \sum_{k\in\mathbb{Z}_p^2}\phi_l(k)\rho^{k\cdot x}$; the sign-flipped coefficient sits at the character $\chi_{l,T} := 2l\,\mathbf{1}_T$ with $|\phi_l(\chi_{l,T})| = |\beta_{l,T}|$ (the computation of the companion note's Lemma 11 uses no property of the inputs). Let $\hat g_\delta(k)$ be the Fourier coefficients of $x \mapsto g(x,\delta)$, let $\delta^\star$ minimise $\sum_l a_l\kappa_l(\delta)$, and let $L' = \{l : |\beta_l| > a_l\}$. Suppose there are T_l ($l \in L'$) with $|\beta_{l,T_l}| = \max_T|\beta_{l,T}|$ such that $\Lambda = \{\chi_{l,T_l}\}_{l\in L'}$ is*

- *(D)* dissociated*: the $3^{|L'|}$ sums $\sum_l \sigma_l\chi_{l,T_l}$, $\sigma \in \{0,\pm1\}^{L'}$, are pairwise distinct (no relation with coefficients in $\{0,\pm1,\pm2\}$); and*
- *(I)* isolated *at $\pm\delta^\star$: $\hat g_{\pm\delta^\star}$ vanishes on the sums $\sum_l \sigma_l\chi_{l,T_l}$ with $|\operatorname{spt}\sigma| \ge 2$, and at $\pm\chi_{l,T_l}$ only feature l's own sign-flipped mode contributes.*

Then $\gamma(\theta) \le 3^{1-s}B_p\Gamma_s(p)$, strictly if $L' \neq \emptyset$. Hence every global maximiser either has no cheap feature, and then uses exactly $K^*(s,p)$ features (Proposition 18), or its cheap features violate (D) or (I).

Proof. For any $\nu \ge 0$ with $\mathbb{E}\nu = 1$, $h \le \mathbb{E}_x[\nu(x)g(x,\delta)]$. Take $\nu = \prod_{l\in L'}\big(1-\cos(2\pi\langle\chi_{l,T_l},x\rangle/p+\psi_l)\big)$ with $\psi_l = \arg\hat g_\delta(\chi_{l,T_l})$. By (D) every character of ν has a unique representation, so $\hat\nu(0) = 1$ and $\hat\nu(\pm\chi_{l,T_l}) = -\frac12 e^{\pm i\psi_l}$, and $\hat\nu$ is supported on $\{\sum\sigma_l\chi_{l,T_l}\}$. By (I),

$$h \le \hat g_\delta(0) - \sum_{l\in L'}|\hat g_\delta(\chi_{l,T_l})| = \mathbb{E}_x g(\cdot,\delta) - \sum_{l\in L'}|\beta_{l,T_l}|\sqrt{\kappa_l(\delta)/2},$$

because $|\hat g_\delta(\chi_{l,T_l})| = \frac12|\beta_{l,T_l}||1-\rho^{-l\delta}|$. Average this over $\delta = \pm\delta^\star$ and use Part I's Theorem A for $l \in L'$ ($|\beta_l| - a_l \le R_l/(2\sqrt3) \le \frac12|\beta_{l,T_l}|$; Lemma 6 of the paper makes it valid for any inputs):

$$h \le \sum_l a_l\kappa_l(\delta^\star) + \sum_{l\in L'}|\beta_{l,T_l}|\Big(\tfrac12\kappa_l(\delta^\star) - \sqrt{\kappa_l(\delta^\star)/2}\Big).$$

Each bracket is negative because $\kappa_l < 2$, and $\beta_{l,T_l} \neq 0$ for $l \in L'$ (the paper's Lemma 7). Finally $\sum_l a_l\kappa_l(\delta^\star) \le 3^{1-s}(C_bP)^s\Gamma_s$ as before. For the last sentence, $\gamma(\theta^\star) \ge 3^{1-s}B_p\Gamma_s$. □

Two remarks on what Theorem 19 says. (i) It is pointwise, needs no circularity, and holds at $p = 3$. There the counterexamples violate (I): with constant input components, the mode $\mathbf{1}_T$ coincides with $-\chi_{1,T}$ because $2 \equiv -1 \pmod 3$. This is the non-circularity of the companion note, seen as a spectral coincidence. (ii) It replaces the $\sqrt{|L'|}$ loss of Theorem 17 by nothing. Collective undercutting, and compensation by other fields, are possible *only* through exact arithmetic: an additive relation $\sum\sigma_l\,2l\mathbf{1}_{T_l} = 0$ among the cheap features, or a mode of some field landing exactly on a cheap feature's sign-flipped character (for instance an input frequency $3l$, whose mode is the conjugate partner of $\chi_{l,T}$). In class $\mathcal{C}$ the second cannot happen, and (I) reduces to the absence of relations $\sum\sigma_l\chi_{l,T_l} \in \{\pm\chi_{k,T}\}$. The programme supports have $K^* = 2, 4$ at $p = 11, 31$ ($b = 2$;

exhaustive vertex enumeration at $p = 11$, and at $p = 31$ for supports up to size 5) and $K^* \approx 6$–7 at $p = 101$ (heuristic search only; the programme is a hard concave minimisation there), i.e. $K^* \approx 1.3$–$1.5 \ln p$, below the capacity $2 \log_3 p \approx 1.8 \ln p$ of dissociated sets in $\mathbb{Z}_p^2$. A dissociated choice of T_l exists for the programme supports at $p = 31$ and 211, but none was found at $p = 101$ or 401. Moreover the network, not the proof, decides which T carries a feature's interference. With all interference on $T = \{1, 2\}$ the characters are 1-dimensional, and $\log_3 p \approx 0.9 \ln p < K^*$ forces relations, so exact dissociation cannot cover every feature being cheap at large p. There one has to use $\nu = \prod(1 - \varepsilon \cos)$ with $\varepsilon < 1$, whose relation terms carry weight $(\varepsilon/2)^{\text{length}}$. The room for that is $1/2$ against the sharp single-feature constant $c_1(2, s) = \sup(|\beta| - a)/\max_T |\beta_T|$, which equals $1 - 3^{1-s}$ (Theorem 34; a single neuron; 0.423 at $b = 2$) and so is strictly below $1/2$.

Theorem 20 (the price of a feature at $n = 2$, every prime, arbitrary inputs)**.** *For $n = 2$, $2 \le b \le 3$ and every odd prime p, every single-feature network (outputs at frequency 1, inputs arbitrary, any width) satisfies*

$$\min_x \operatorname{cm}(\Phi_1(x)) \;\le\; \operatorname{Re}\beta_1 - \cos(\pi/p) \max_T |\beta_{1,T}| \;\le\; 3^{1-s} Q^s \;\le\; 3^{1-s} B_p \|\theta\|_{2,b}^{\nu}.$$

Hence $\pi_{2,b}(p) = 3^{1-s} = \nu^{1-s}$: a single feature cannot be bought below the interference-free price at $n = 2$, at any prime. This settles (Q1) for $n = 2$, and with the price–programme theorem the decoupled value is the interference-free value. At $p = 3$ it recovers Part I.

Proof. Fix $T \neq \emptyset$ and $\chi = \chi_{1,T} = 2\mathbf{1}_T$. A mode of Φ_1 at $\pm\chi$ needs input indices k_t with $k_t + 1 = \pm 2$ on T and $k_t + 1 = 0$ off T. So the only modes there are the sign-flipped mode, with coefficient a, $|a| = |\beta_{1,T}|$, and the conjugate partner (input frequency 3 on T), with coefficient a'; this holds for arbitrary inputs. Put $b = \overline{a'}$ and $c = 1 - \rho^{-1} = 2 \sin(\pi/p) e^{i\alpha}$, $\alpha = \frac{\pi}{2} - \frac{\pi}{p}$. For $\delta = \pm 1$ the gap $g_\delta = \operatorname{Re}(\Phi_1 c_\delta)$, with $c_{\pm 1} = c, \bar{c}$, has coefficient $\hat{g}_\delta(\chi) = \frac{1}{2}(a c_\delta + b \bar{c}_\delta)$. With $\nu = 1 - \cos(2\pi \langle \chi, x \rangle / p + \arg \hat{g}_\delta(\chi))$, $h \le \mathbb{E}[\nu g_\delta] = \mathbb{E} g_\delta - |\hat{g}_\delta(\chi)|$. Averaging over $\delta = \pm 1$, $\frac{1}{2}(\mathbb{E} g_1 + \mathbb{E} g_{-1}) = \kappa_{\min} \operatorname{Re}\beta_1$. Moreover writing $ac + b\bar{c} = |c| X$ and $a\bar{c} + bc = |c| Y$, we have $X e^{i\alpha} - Y e^{-i\alpha} = 2ia \sin 2\alpha$ (the b-terms cancel), so $|ac + b\bar{c}| + |a\bar{c} + bc| \ge 2|a||c| |\sin 2\alpha|$. So

$$h \le \kappa_{\min} \operatorname{Re}\beta_1 - \tfrac{1}{2}|a| \cdot 2 \sin(\tfrac{\pi}{p}) \sin(\tfrac{2\pi}{p}) = \kappa_{\min} \big(\operatorname{Re}\beta_1 - \cos(\tfrac{\pi}{p}) |\beta_{1,T}| \big),$$

and $h = \kappa_{\min} \min_x \operatorname{cm}$ (companion note, Lemma 3). Take T maximising $|\beta_{1,T}|$. Part I's Theorem A (valid for arbitrary inputs by the paper's Lemma 6) gives $\operatorname{Re}\beta_1 \le 3^{1-s} Q^s + \frac{1}{2} \max_T |\beta_{1,T}|$, and $\cos(\pi/p) \ge \frac{1}{2}$. Finally $Q = \sum_i |\beta_{i1}|^r \le C_b P$. □

The partner, which is the $p = 3$ mechanism's lever (there $3 \equiv 0$, i.e. the constant input component), cannot help: it can cancel the sign-flipped coefficient at one edge, but not at both, and the residual is the factor $\cos(\pi/p)$. For several features the same averaging over $\pm\delta^\star$ gives the gain $\kappa_l(\delta^\star) |\cos(\pi l \delta^\star/p)| \, |\beta_{l,T_l}|$ per feature when only its own partner interferes, so Theorem 19 survives own partners whenever $|\cos(\pi l \delta^\star/p)| \ge \frac{1}{2}$ for the cheap l.

Every n. The proof of Theorem 19 uses $n = 2$ only through Theorem A. For general n, give each cheap feature l a family $\mathcal{T}_l$ of non-empty $T \subseteq [n]$ with linearly independent indicators (their characters $\chi_{l,T}$ are then dissociated for $p > 2n$), and take the Riesz product over all $\chi_{l,T}$, $T \in \mathcal{T}_l$, $l \in L'$. The same computation gives $\gamma(\theta) \le \nu^{1-s} B_p \Gamma_s(p)$ (strictly if $L' \neq \emptyset$) under (D), (I) for this larger set, provided

$$(\mathrm{H}_n) \qquad |\beta_l| - \nu^{1-s} Q_l^s \;\le\; \tfrac{1}{2} \textstyle\sum_{T \in \mathcal{T}_l} |\beta_{l,T}| \quad \text{for some admissible } \mathcal{T}_l.$$

(H_n) is a single-feature statement about atoms $\beta_i \in \mathbb{C}$, $\xi_i \in \mathbb{T}^n$ (the paper's Lemma 6). The families must be *collision-free*: no signed sum of ≥ 2 members' indicators equals $\pm\mathbf{1}_{T'}$, otherwise the feature's own coefficient at $\chi_{l,T'}$ enters with uncontrolled phase. For $n = 2$ only single sets qualify, and Theorem A gives $c_1 \leq \frac{1}{2}$ on all of $b \in [2, 3]$. For $n \geq 3$, (H_n) at $\frac{1}{2}$ is false in the relevant range (computed, with certified examples).

$n = 3$: two atoms $\beta = e^{\mp i\varphi}/(2\cos\varphi)$, $\xi_2 = e^{i(\pi/2-\varphi)}(1, 1, 1)$ kill every $|T| = 2$ coefficient and give $c_1^{\mathrm{cf}}(3, s) \geq (\cos\varphi - 2^{1-s})/\cos(\frac{\pi}{4} + \frac{\varphi}{2})$. This exceeds $\frac{1}{2}$ for $b < 2.51$ and equals 0.767 at $b = 2$, still below $\cos(\pi/5) = 0.809$.

$n = 4$: three atoms $d, ce^{\pm i\psi}$, $\xi = (e^{\pm i\theta}, \dots)$ (certified at 50 digits over all 80 collision-free families) give $c_1^{\mathrm{cf}}(4, \frac{5}{2}) \geq 1.127 > 1$. So at $n = 4$, $b = 2$ a feature's own sign-flipped characters cannot pay for its bonus pointwise at any prime.

So the pointwise (Riesz) route closes the single-feature question for $n = 2$ (Theorem 20), is consistent with it for $n = 3$ at every $p \geq 5$ (threshold $\cos(\pi/p)$, unproved), and cannot work for $n \geq 4$ at $b = 2$. There the L^2 (circularity) route of the companion note's Theorem 14 is needed. It is rigorous for single-frequency inputs whenever $k^*(n, s) \leq k_p$, but k^* is known only numerically ($0.35, 0.54, 0.67, 0.76, 0.93$ for $n = 2, 3, 4, 5, 7$ at $b = 2$, against $k_5 = 0.946$). Files: `mechanism_scripts/Hn/`.

Proposition 21 (joint L^2/pointwise bound for single features, $p \geq 5$)**.** *Let $p \geq 5$ and let θ be a single-frequency single-feature network with mean-zero inputs (every u_{it} and w_i in* $\mathrm{span}\{\cos, \sin\}$ *of frequency* 1*). Put $\sigma^2 = \frac{1}{2}\sum_{T\neq\emptyset} |\beta_{1,T}|^2$. Then*

$$\min_x \mathrm{cm}(\Phi_1) \;\leq\; \mathrm{Re}\,\beta_1 - \max\Big(k_p\,\sigma,\; \cos(\tfrac{\pi}{p}) \max_{\mathcal{T}} \textstyle\sum_{T\in\mathcal{T}} |\beta_{1,T}|\Big),$$

with $\mathcal{T}$ ranging over collision-free families. Hence θ does not beat the interference-free single-feature value whenever the joint constant $J(n, s, p) := \sup(|A| - \nu^{1-s}Q^s)/\max(k_p\sigma, \cos(\frac{\pi}{p})\max_{\mathcal{T}}\sum|\beta_T|)$ (over atom systems) is at most 1*.*

Proof. The first term is the companion note's Theorem 14 (circularity holds for single-frequency fields at $p \geq 5$). For the second, run the proof of Theorem 20 with the Riesz product over the family: with mean-zero inputs every mode of Φ_1 has per-slot characters in $\{0, 2\}$, so a multi-sum $\sum \sigma_T\chi_T$ carries a mode only if it is $\pm 2\mathbf{1}_{T'}$, which collision-freeness excludes. □

Local search gives $J(3, 2, 5) \approx 0.57$ and $J(4, \frac{5}{2}, 5) \approx 0.71$ ($b = 2$). For $n \leq 7$ the L^2 term alone already suffices numerically ($k^* < k_5$). The joint constant matters from $n = 8$, where $k^*(8, \frac{9}{2}) \geq 0.99 > k_5$. These are lower bounds on the suprema from local search, so they are evidence, not proof. A direct search at $p = 5$, $n = 8$ (single-frequency inputs, seeded at the IF block) found nothing above the IF value, and no feasible network of width $8 < \nu$.

Room in Theorem 19. The factor $1 - \cos\theta$ extracts half of each character's amplitude. A shifted Fejér kernel $F_d(\theta + \pi) = 1 + 2\sum_{k=1}^{d}(1 - \frac{k}{d+1})\cos k(\theta + \pi) \geq 0$ extracts the fraction $\frac{d}{d+1}$. The proof is the same, provided the representations $\sum_l k_l\chi_{l,T_l}$ with $|k_l| \leq d$ are unique and $\hat{g}$ vanishes at the multiples $k\chi_{l,T_l}$, $2 \leq |k| \leq d$. In class $\mathcal{C}$ the latter fails only if some feature is a multiple kl of a cheap feature l. The conclusion then holds whenever $|\beta_l| - a_l \leq \frac{d}{d+1}\max_T |\beta_{l,T}|$, which for $n = 2$ leaves a factor $\frac{2d}{d+1}$ of room over Theorem A, and more against the sharp constant $1 - 3^{1-s}$. That room is what a weighted product $\prod(1 - \varepsilon\cos)$ needs to absorb the relations that dissociation cannot exclude at large p. The price is a smaller capacity: unique representations allow at most $n\log_{2d+1} p$ characters.

Proposition 22 (hiding a character needs many modes; any network)**.** *For $\chi \in \mathbb{Z}_p^n \setminus \{0\}$ let $\phi_j(\chi)$ be the coefficient of Φ_j at $\rho^{\langle \chi, x\rangle}$ and $F_\chi(\delta) := \mathbb{E}_x[\sigma(x,\delta)\rho^{-\langle \chi,x\rangle}]$ ($\delta \neq 0$), $F_\chi(0) := 0$. Then $\phi_j(\chi) = -\frac{2}{p}\sum_\delta \rho^{j\delta} F_\chi(\delta)$ for every $j \in F$, and $\phi_j(-\chi) = -\frac{2}{p}\overline{\sum_\delta \rho^{-j\delta} F_\chi(\delta)}$. Consequently, if some field has a mode at $\pm\chi$ and F_χ vanishes on a symmetric set $B \subseteq \mathbb{Z}_p \setminus \{0\}$, then at least $|B|+1$ of the $p-1$ coefficients $\{\phi_j(\pm\chi)\}_{j\in F}$ are non-zero.*

Proof. The first formula is the χ-coefficient of the slack identity (Lemma 11); the second follows because σ is real. So $\hat{F}_\chi(j) = \sum_\delta F_\chi(\delta)\rho^{j\delta}$ equals $-\frac{p}{2}\phi_j(\chi)$ for $j \in F$ and $-\frac{p}{2}\overline{\phi_{-j}(-\chi)}$ for $j \in -F$. $F_\chi \not\equiv 0$ and $|\operatorname{spt} F_\chi| \leq p-1-|B|$, so Tao's uncertainty principle for $\mathbb{Z}_p$ ($|\operatorname{spt} f| + |\operatorname{spt} \hat{f}| \geq p+1$) gives $|\operatorname{spt} \hat{F}_\chi| \geq |B|+2$, of which at most one is $j=0$. □

For $\chi = \chi_{l,T}$ every mode $(j, \pm\chi)$ other than feature l's own needs input energy at an off-frequency ($2l-j$ or $-2l-j$ on T). So hiding one sign-flipped character from the $|B| \geq 2K^*$ binding directions of a programme vertex needs at least $2K^*$ off-frequency mode families. And by the paper's Lemma 6 each such family comes with 2^n-1 sign-flipped siblings of equal size at other characters, which must in turn be paid for or hidden. The proposition is exact only where the slack vanishes identically. Turning it into the quantitative statement the compensation case needs (a robust uncertainty principle on $\mathbb{Z}_p$, combined with this sibling conservation) is the remaining analytic step.

The relations case is a one-sided Sidon problem. Suppose all interference sits on $T = \{1,2\}$, the case in which the network can force relations. Then the conditional gap is $M_\delta + \varphi(w) - \varphi(w-\delta)$ with $\varphi(w) = \sum_l A_l \cos(2\pi l w/p + \theta_l)$ (for two aligned neurons per feature this is exact; see the relations note in `mechanism_scripts`). Evaluating every direction at one common point w_0 and averaging with the programme's dual weights μ (symmetric, on the binding set) gives

$$h \;\leq\; \sum_\delta \mu_\delta M_\delta - \Big(-\min_w \psi_\mu(w)\Big), \qquad \psi_\mu(w) = \sum_l A_l K_l(\mu)\cos(2\pi l w/p + \theta_l),$$

because $\varphi - \mu * \varphi$ has coefficients $A_l K_l(\mu)$. So the interference-free bound for this family follows from the one-sided Sidon inequality $-\min\psi_\mu \geq c\sum_l A_l K_l(\mu)$ with $c = 1-(2/3)^{s-1}$ (0.18 at $b=2$), for adversarial phases. That inequality holds with $c = \frac{1}{2}$ for dissociated S (Riesz), and numerically with $c \approx 0.8$ for the programme supports. It fails for arithmetic progressions, whose Rudin–Shapiro polynomials are flat, but those have poor coverage. What is missing is a "coverage implies one-sided Sidon" lemma for $|S| \approx 1.5\ln p$, which is Chowla's cosine problem with phases and with a coverage hypothesis.

Spurious features need all-slot content. A clean amplitude at q needs a neuron with frequency-q content in every input slot and an output at q. So a cheap single-feature mechanism built from single-frequency inputs, even if it beat the interference-free value, would not change the count: compensating outputs at other frequencies carry no clean amplitude. The count can change only through (i) a shift of the allocation's support or (ii) neurons with multi-frequency inputs in all slots (sign-vector type mechanisms).

Parasitic orbits: compensation is cheap, but does not pay in the models tested. An interference-free block for feature j can carry a hidden compensating mode. Give its three neurons input components at the off-frequency $q = 2l-j$ in slot 1, with relative size ρ and phases $w_i = e^{i\arg \hat{u}_{i1}(q)}$. With the paper's block phases $\alpha_i = \pi(i-1)/3$, the three sibling modes

$(-q,-j),(q,j),(-q,j)$ cancel iff $w_1 = w_2 e^{i\pi/3}$ and $w_3 = w_2 e^{-i\pi/3}$. The surviving mode lands on $\chi_{l,\{1\}} = (2l,0)$ with size exactly $\rho\, a_j/\sqrt{1+\rho^2}$, while the block's clean amplitude drops to $a_j/\sqrt{1+\rho^2}$ at fixed cost. So a compensating mode of size M costs only $O(M^2)$. We tested the most favourable configuration (one feature on two aligned neurons, interference on the single character $(2l,0)$, all other features carrying parasitic orbits, all phases, sizes and the allocation optimised). In the large-p limit the margin is exactly $\min_\delta[\sum_k a_k\kappa_k/\sqrt{1+\rho_k^2} + (\frac{3}{2})^{s-1}a_l\kappa_l - |Z(\delta)|]$, where Z is the total coefficient at $(2l,0)$. When the allocation is left free, the optimiser always switches the cheap feature off. With the cheap feature forced to keep $1, \frac{1}{2}, \frac{1}{5}$ of its programme share at $p = 31$ $(b = 2)$, the best margins are 0.49–0.64, 0.60–0.75 and 0.67–0.82 of the IF value (over the four choices of cheap feature). They rise monotonically back towards 1 as the share shrinks. (An earlier run reported gains; they came entirely from a suboptimal programme solver, since corrected.) By Proposition 22, exact hiding from a programme vertex's $2K^*$ binding directions needs more modes than the features' outputs provide.

Remark 23 (why the remaining step is a question of constants)**.** Let K features each carry interference of size ε (relative to their amplitudes). Their total gain at a binding direction is at most $k^*\varepsilon\sum_l \kappa_l(\delta^\star) \asymp k^*\varepsilon K$. The slack they force is at least $\varepsilon\sqrt{\sum_l \kappa_l(\delta^\star)/(2^n-1)} \asymp \varepsilon\sqrt{K}$ (proof of Theorem 17). This is also the right order for an upper bound: translating each feature's inputs rotates its sign-flipped phases independently, and a sum of K cosines with independent phases has supremum $O(\sqrt{K\log p})$. At the programme's own scale $K = K^* = \Theta(\log p)$ (paper, Theorem 11), gain and forced slack are therefore of the same order, uniformly in p. So collective undercutting is neither negligible nor dominant at any prime, and no order-of-magnitude argument can settle the count. A proof has to compare three sharp constants: the true $k^*(n,s)$ (at least $c_n(s-1)$, from $\nu-1$ equal aligned atoms), the best cancellation constant for the profiles φ_T under the joint pointwise constraint, and the coverage constant of the programme at its binding directions. The experiments at $p = 5, 7$ (no compensated network within $4\times$ of IF) say the first is small against the other two at those primes. Theorem 19 is such a comparison, and it is won by the sharp pair $\frac{1}{2}$ (Riesz gain per feature) against $c_1(2,s) \le \frac{1}{2}$ (Theorem A) whenever the cheap features' characters are dissociated. A dissociated set in $\mathbb{Z}_p^n$ has at most $n\log_3 p$ elements (3^K distinct sums in a group of order p^n). For $n = 2$, $b = 2$ that is $\approx 1.8\ln p$, against $K^* \approx 1.3$–$1.5\ln p$, so exact dissociation is in reach in principle. But an adversarial choice of T (all interference on $T = \{1,2\}$) reduces the capacity to $\log_3 p \approx 0.9\ln p$, which is below K^*. A direct test agrees that phases cannot flatten the profiles: realising every feature of a 6-feature allocation at $p = 31$ by a two-neuron sub-network and optimising all translations gives margin $-0.47\times$ the IF margin, not the > 1 needed. Each feature's field is a product polynomial $P_l(\zeta^l)$, whose coherent excursions phases do not remove.

9 Architecture of a proof of the count

Everything proved above, and everything still open, is an instance of one inequality. Throughout this section θ is an arbitrary network (any n, p, b, width and inputs), $P = \|\theta\|_{2,b}^b$, $a_l = \nu^{1-s}Q_l^s$, $\Delta(\theta) = \arg\min_{\delta\neq 0}\sum_l a_l\kappa_l(\delta)$, $\mathcal{M}(\theta)$ is the set of symmetric probability measures on $\Delta(\theta)$, and $K_l(\mu) = \sum_\delta \mu_\delta\kappa_l(\delta)$. Two non-negative accounts measure how far the allocation is from the interference-free optimum:

$$U(\theta) = \nu^{1-s}\Big((C_bP)^s - \big(\textstyle\sum_l Q_l\big)^s\Big)\Gamma_s, \qquad G(\theta) = \nu^{1-s}\big(\textstyle\sum_l Q_l\big)^s\,\Gamma_s - \min_{\delta\neq 0}\sum_l a_l\kappa_l(\delta).$$

The *unused budget* U collects norm that buys no clean amplitude: off-frequency input content, unbalanced or constant components, and at $b > 2$ multi-frequency neurons. The *coverage gap*

G vanishes iff $(Q_l^s)_l$ is proportional to a maximiser of Γ_s. For a density $w \geq 0$ on $\mathbb{Z}_p^n$ with $\mathbb{E}w = 1$ let $D_\delta(w) = \mathbb{E}_x g(x,\delta) - \mathbb{E}_x[w(x)g(x,\delta)]$ be the slack that w detects in direction δ, and let $L'(\theta) = \{l : |\beta_l| > a_l\}$ be the cheap features.

Proposition 24 (master inequality)**.** *For every $\mu \in \mathcal{M}(\theta)$ and all densities $(w_\delta)_{\delta \neq 0}$,*

$$h(\theta) \;\leq\; \nu^{1-s} B_p \Gamma_s(p)\, \|\theta\|_{2,b}^{\nu} \;+\; \sum_l (\operatorname{Re}\beta_l - a_l) K_l(\mu) \;-\; \sum_\delta \mu_\delta D_\delta(w_\delta) \;-\; U(\theta) - G(\theta).$$

So $\gamma(\theta) \leq \nu^{1-s} B_p \Gamma_s(p)$ as soon as

$$\text{(MI)} \qquad \sum_l (\operatorname{Re}\beta_l - a_l) K_l(\mu) \;\leq\; \sum_\delta \mu_\delta D_\delta(w_\delta) + U(\theta) + G(\theta)$$

for some $\mu \in \mathcal{M}(\theta)$ and some tests (w_δ), and the inequality is strict if (MI) is. If (MI) holds strictly for every network with $L'(\theta) \neq \emptyset$, then every global maximiser of the $L_{2,b}$-margin uses exactly $|\operatorname{spt} y^\star|$ features for some maximiser $y^\star$ of $\Gamma_s(p)$.

Proof. For each δ, $h \leq \mathbb{E}[w_\delta g(\cdot,\delta)] = \mathbb{E}g(\cdot,\delta) - D_\delta(w_\delta)$. Average with μ. Since μ is symmetric, $\sum_\delta \mu_\delta \mathbb{E}g(\cdot,\delta) = \sum_l \operatorname{Re}\beta_l\, K_l(\mu)$. Since μ lives on $\Delta(\theta)$, $\sum_l a_l K_l(\mu) = \min_\delta \sum_l a_l \kappa_l(\delta) = \nu^{1-s}(C_b P)^s \Gamma_s - U - G$, and $(C_b P)^s = B_p \|\theta\|_{2,b}^{\nu}$. A global maximiser $\theta^\star$ has $\gamma(\theta^\star) \geq \nu^{1-s} B_p \Gamma_s$ (paper, Theorem 8), so (MI) cannot hold strictly for it. Hence $L'(\theta^\star) = \emptyset$, and Proposition 18 applies. □

Point masses at the minimisers of $g(\cdot,\delta)$ are admissible tests, so (MI) loses nothing beyond the choice of μ. The count, for all n, p, b, is therefore the single statement that *a cheap feature is always strictly paid for*, by detected slack, unused budget or coverage gap.

Two factors. Every argument in Section 8 proves (MI) in the same way. For each $l \in L'$ it chooses characters $X_l \subseteq \mathbb{Z}_p^n \setminus \{0\}$ and weights $\lambda_{l,\chi} > 0$, and then proves two inequalities that meet in the middle:

(P) *payment*, one frequency at a time: $\operatorname{Re}\beta_l - a_l \leq \sum_{\chi \in X_l} \lambda_{l,\chi} |\beta_{l,\chi}|$, where $\beta_{l,\chi}$ is the coefficient that feature l's own atoms put at χ (the sign-flipped coefficient when $\chi = \chi_{l,T}$);

(A) *detection*, all frequencies together: $\sum_\delta \mu_\delta D_\delta(w_\delta) + U + G \geq \sum_{l \in L'} K_l(\mu) \sum_{\chi \in X_l} \lambda_{l,\chi} |\beta_{l,\chi}|$.

By the paper's Lemma 6, (P) is a statement about one finite family of atoms (β_i, ξ_i), the same in every network, so it is decided on single-feature networks. Features interact only in (A). There is also an L^2 pair: (P_2) $\operatorname{Re}\beta_l - a_l \leq k^* \sigma_l$ with $\sigma_l^2 = \frac{1}{2}\sum_T |\beta_{l,T}|^2$, and ($\mathrm{A}_2$), Bhatia–Davis plus Parseval as in the proof of Theorem 17. The results above are instances:

result	(P)	(A)
Prop. 18	vacuous ($L' = \emptyset$)	vacuous
Thm. 20	Theorem A; $X = \{\chi_{1,T}\}$, $\lambda = \frac{1}{2}$	both-edge averaging; own partner allowed
Thm. 19	Theorem A; $X_l = \{\chi_{l,T_l}\}$, $\lambda = \frac{1}{2}$	Riesz product, under (D) and (I)
(H_n)	collision-free families, $\lambda = \frac{1}{2}$; false for $n \geq 3$ in part of the range	Riesz product
Thm. 17	(P_2)	(A_2) in class $\mathcal{C}$; loses $\sqrt{\lvert L'\rvert}$
Prop. 21	max of (P) and (P_2)	single feature

Lemma 25 (the two defects of a Riesz test)**.** *Let $X = \bigcup_{l\in L'} X_l$ with $X_l \subseteq \{\chi_{l,T} : T \neq \emptyset\}$, let $\varepsilon_\chi \in (0,1]$, and suppose no signed sum $\sum_\chi \sigma_\chi \chi$ with $\sigma \in \{0,\pm 1\}^X$, $|\operatorname{spt}\sigma| \geq 2$, vanishes. Put $w = \prod_{\chi\in X}\big(1 - \varepsilon_\chi \cos(2\pi\langle\chi, x\rangle/p + \psi_\chi)\big)$ with $\psi_\chi = \arg\hat{g}_\delta(\chi)$. Then*

$$D_\delta(w) \;\geq\; \sum_{l\in L'}\sum_{\chi_{l,T}\in X_l} \varepsilon_{\chi_{l,T}}\Big(|\beta_{l,T}|\sqrt{\kappa_l(\delta)/2} - |Z_{l,T}(\delta)|\Big) \;-\; \sum_{|\operatorname{spt}\sigma|\geq 2}\;\prod_{\sigma_\chi\neq 0}\frac{\varepsilon_\chi}{2}\;\Big|\hat{g}_\delta\Big(\sum_\chi \sigma_\chi\chi\Big)\Big|,$$

where $Z_{l,T}(\delta)$ is everything in $\hat{g}_\delta(\chi_{l,T})$ except the contribution of feature l's own sign-flipped mode.

Proof. $w \geq 0$. Expanding the product gives $\hat{w}(0) = 1$ (no signed sum vanishes), coefficients $-\frac{\varepsilon_\chi}{2}e^{\pm i\psi_\chi}$ at $\pm\chi$, and coefficients of modulus $\prod_{\sigma_\chi\neq 0}\frac{\varepsilon_\chi}{2}$ at the other signed sums. So $\mathbb{E}[w g_\delta] \leq \mathbb{E}g_\delta - \sum_\chi \varepsilon_\chi|\hat{g}_\delta(\chi)| + \sum_{|\operatorname{spt}\sigma|\geq 2}\prod\frac{\varepsilon_\chi}{2}|\hat{g}_\delta(\sum\sigma_\chi\chi)|$. Finally $g_\delta = \sum_j \operatorname{Re}(\Phi_j c_j(\delta))$ puts $\frac{1}{2}\phi_l(\chi_{l,T})c_l(\delta)$ at $\chi_{l,T}$ from feature l's sign-flipped mode, of modulus $\frac{1}{2}|\beta_{l,T}|\sqrt{2\kappa_l(\delta)}$, and the triangle inequality gives the rest. □

Without either defect the lemma gives (A) with $\lambda = \varepsilon/\sqrt{2\kappa_l} \geq \frac{1}{2}$ at $\varepsilon = 1$; this is Theorem 19. So (A) can fail in only two ways:

(R) *relations*: a signed sum of ≥ 2 chosen characters vanishes, or lands where $\hat{g}_\delta \neq 0$;

(C) *compensation*: a mode other than feature l's own sign-flipped mode sits at $\chi_{l,T}$.

(O1) is (R) and (O2) is (C). The next two lemmas say where (C) can come from and what it costs.

Lemma 26 (compensation channels; any n)**.** *Let $\phi_j(\chi) = 2p^{-\nu}\sum_i \hat{w}_i(j)\prod_t \hat{u}_{it}(\chi_t - j)$ be the coefficient of Φ_j at χ. Every contribution to $\hat{g}_\delta(\chi_{l,T})$ other than $\phi_l(\chi_{l,T})$ comes from a neuron with an output frequency j that has, in every slot $t \in T$, input content at a frequency $q \notin \{\pm j\}$, and on-frequency content $\pm j$ in the slots off T. Either*

(i) $j = l$ and $q = \pm 3l$: the own partner*, through $\phi_l(-\chi_{l,T})$; or*

(ii) $j \neq l$ and $q = \pm(2l - j)$ or $q = \pm(2l + j)$: a foreign mode*.*

In particular nothing compensates in class $\mathcal{C}$.

Proof. $\hat{g}_\delta(\chi) = \frac{1}{2}\sum_j\big(\phi_j(\chi)c_j(\delta) + \overline{\phi_j(-\chi)c_j(\delta)}\big)$. A term of $\phi_j(\pm\chi_{l,T})$ needs $\hat{u}_{it}(\pm 2l - j) \neq 0$ for $t \in T$ and $\hat{u}_{it}(-j) \neq 0$ for $t \notin T$. Since $l, j \in F$ and p is odd, $\pm 2l - j \equiv \pm j$ only for $l = j$ with the sign $+$, which is the own mode. For $j = l$ the other sign gives $-3l$. Real inputs have $|\hat{u}(q)| = |\hat{u}(-q)|$. □

Lemma 27 (paid compensation is quadratic)**.** *Let ω be a neuron, j one of its output frequencies, and C the coefficient of its frequency-j field at a character reached through an input frequency $q \notin \{0, \pm j\}$ on the slots of $T \neq \emptyset$ and $\pm j$ off T. If q is not an output frequency of ω ($\hat{w}(q) = 0$), then*

$$\sum_{k\neq j}|\beta_k(\omega)|^r + \big(|\beta_j(\omega)|^2 + |C|^2\big)^{r/2} \;\leq\; C_b\|\omega\|_2^b.$$

For a single-output neuron this reads $|\beta_j(\omega)|^2 + |C|^2 \leq e(\omega)^2$, $e(\omega) = B_p\|\omega\|_2^\nu$, the clean amplitude its norm buys. If neurons i of this kind carry modes C_i at a common character, with total $M = \sum_i C_i$, then $\sum_i|\beta_j(\omega_i)| \leq \sqrt{(\sum_i e(\omega_i))^2 - |M|^2}$. The parasitic orbits above attain these bounds.

Proof. Let $m_k = |\hat{w}(k)|^2 + \sum_t |\hat{u}_t(k)|^2$ for $k \in F$ and $\tau = \sum_{t\in T} |\hat{u}_t(q)|^2 \le m_q$. With real inputs, $|\beta_j(\omega)| = 2p^{-\nu}|\hat{w}(j)| \prod_t |\hat{u}_t(j)|$ and $|C| = 2p^{-\nu}|\hat{w}(j)| \prod_{t\in T} |\hat{u}_t(q)| \prod_{t\notin T} |\hat{u}_t(j)|$, so

$$|\beta_j(\omega)|^2 + |C|^2 \le (2p^{-\nu})^2 |\hat{w}(j)|^2 \prod_{t\notin T} |\hat{u}_t(j)|^2 \prod_{t\in T} \big(|\hat{u}_t(j)|^2 + |\hat{u}_t(q)|^2\big) \le (2p^{-\nu})^2 \Big(\frac{m_j + \tau}{\nu}\Big)^\nu$$

by AM–GM. Now run the paper's proof of the single-neuron budget (Appendix C) with m_j replaced by $m_j + \tau$ and m_q by $m_q - \tau$; the total $\sum_k m_k \le \frac{p}{2}\|\omega\|_2^2$ is unchanged, and $\beta_q(\omega) = 0$ because $\hat{w}(q) = 0$. For one output frequency, $|\hat{w}(j)|^2 \le \frac{p}{2}\|w\|^2$ and AM–GM over the ν norms give $e(\omega)$ directly. The last claim is Aczél's inequality $\sum_i \sqrt{e_i^2 - |C_i|^2} \le \sqrt{(\sum_i e_i)^2 - |\sum_i C_i|^2}$ for $e_i \ge |C_i|$ (reverse Minkowski for timelike vectors). The parasitic orbit has clean amplitude $a_j/\sqrt{1+\rho^2}$ and mode $\rho a_j/\sqrt{1+\rho^2}$, which gives equality. □

So a compensating mode of size M that rides on a feature's block costs that block at least $E - \sqrt{E^2 - M^2} \ge M^2/(2E)$ of clean amplitude. In (MI) this cost is collected by U, because Q_j counts clean amplitudes while the norm is spent.

More generally, the mode is a product of ν factors: $\hat{w}(j)$, the inputs $\hat{u}_t(q)$ on T, and the inputs $\hat{u}_t(j)$ off T. A factor is *paid* if the same neuron uses that mass for a clean amplitude: the j-factors if j is a clean frequency of the neuron, the q-factors if q is. Every unpaid factor is norm that buys no clean amplitude, so it is charged to U. If exactly one factor is unpaid, the cost is quadratic in the mode. Examples are riding (the q-inputs; Lemma 27) and a stray output $\hat{w}(j)$ on a neuron whose inputs carry a clean frequency q with $T = [n]$, which is the cross-feature compensation of Section 6. If several factors are unpaid, the cost is a lower power, down to $|C|^r$ for a dedicated neuron. Compensation is free of norm only in two cases:

(F1) the *own partner* ($q = \pm 3l$ in feature l's own neurons). Both-edge averaging (Theorem 20) leaves the factor $|\cos(\pi l\delta/p)|$ of the sign-flipped coefficient in direction $\pm\delta$, whatever the partner;

(F2) a *shared neuron*, in which both j and q are clean frequencies, so that every factor is paid. At $b = 2$ the budget does not charge for several frequencies in one neuron. If such a clean amplitude does not survive in the network's β_q (or β_j), that feature is bought above price and pays in (MI). So free shared compensation at $\chi_{l,T}$ needs two features $j \ne q$ of one neuron with $2l \equiv \pm j \pm q \pmod p$. For $j = l$ this means that $\pm 3l$ is also a feature. For $b > 2$ shared neurons pay strictly (superadditivity of $t^{b/2}$ is strict).

For $j \ne l$, condition (F2) is a relation $2\cdot(2l) \mp 2j \mp 2q = 0$ with coefficients in $\{0, \pm 1, \pm 2\}$ among doubled frequencies, which is the kind of relation (D) excludes. For $j = l$ it is the relation $3\cdot(2l) = \pm 2q$.

Cost-free compensation defeats Theorem A. Lemma 27 is not optional. Take one cheap feature l of a programme maximiser ($n = 2$, $b = 2$), let the network place its partner and riding modes from the other features of the support *for free*, and let the prover choose μ on B. The value of this game is $V(\lambda) = \min_{b,z,z'} \max_{\pm\delta\in B} \big[\frac{1}{2}(|\hat{g}_\delta(\chi)| + |\hat{g}_{-\delta}(\chi)|) - \lambda\kappa_l(\delta)\big]$ per unit of $|\beta_{l,T}|$, with gain constant λ (Lemma 25, minimax over μ). We computed it for $p \le 31$. With $\lambda = \frac{1}{2}$ (Theorem A) it is 0 at $p = 5, 7$ and negative for some feature at every prime from 11 to 31 (-0.28 at $p = 17$, $l = 8$, where $8\delta^\star \approx p/2$ and $\kappa_l \approx 2$). With the sharp $\lambda = c_1 = 1 - 3^{1-s}$ it is positive for every feature at $p = 5, 7, 13$ and still negative for some feature at $p = 11, 17, 19, 23, 29, 31$. So exact hiding is impossible (Proposition 22), yet near-hiding for free would already beat Theorem A. Any proof of

(O2) has to use the cost of the compensating modes. The costs are quadratic. The gain is at most linear in $|\beta_{l,T}|$, and numerically it is also quadratic in the interference everywhere. A local search over up to six atoms ($n = 2$, $b = 2$) gives $|A| - 3^{1-s}Q^s \le 0.275 \sum_T |\beta_T|^2/|A|$. The extremiser has all $|\beta_T| \approx 0.45|A|$; it is not the IF block and not a single atom, for which the ratio is $c_1/3 = 0.141$. With partner and riding only, the residual of the free game is linear and positive (Proposition 22), so it wins at small interference. With stray outputs it can vanish (Proposition 28 below), and then quadratic gain meets quadratic cost: the decision is a comparison of constants at every size. Files: `mechanism_scripts/freehide.py`.

The single-cheap-feature compensation game, solved. Take one cheap feature l of a programme vertex, its interference resolved by the set T, and every channel of Lemma 26: partner, riding, stray outputs, and dedicated neurons with their siblings. The channels pay their Lorentz costs, and the prover picks μ on B. The value is a *protection curve* $\Pi_T(R)$: the least detected slack plus cost that the network can leave when its sign-flipped mode has size R. By Sion's theorem Π_T has a dual SOCP form. It is convex, and $\Pi_T(R)/R$ is non-decreasing with limit h_T at 0. So a gain bound linear in R (Theorem A's $\frac{1}{2}$, or c_1) wins for the prover iff it is $\le \min_T h_T$.

Proposition 28 (free first-order hiding: a Chebotarev dichotomy)**.** *Let the channels at $\chi_{l,T}$ supply a set M_T of distinct output frequencies $\neq 0, l$, each with a free coefficient. Then $h_T = 0$ (exact first-order hiding from every binding direction) if and only if $|M_T| \ge |B|$. With partner and riding only, $|M_T| = 2|S| - 1 < |B|$ at every vertex, so $h_T > 0$. In the dense case ($S = F$, $B = \mathbb{Z}_p^\times$; e.g. $p = 5, 7$ at $b = 2$) $h_T = \frac{1}{2}$ exactly.*

Proof sketch. $h_T = 0$ iff $c_l|_B$ lies in the span of the $c_m|_B$, $m \in M_T$, i.e. iff the exponentials $\rho^{-k\delta}$, $k \in M_T \cup \{0, l\}$, satisfy a relation on $B \cup \{0\}$ with non-zero coefficient at l. By Chebotarev's theorem every square minor of the Fourier matrix of $\mathbb{Z}_p$ is non-zero. So any $|B| + 1$ of these exponentials are independent on $B \cup \{0\}$, and the relations needed exist iff $|M_T| + 2 \ge |B| + 2$. In the dense case, extend the dual vector u to $\mathbb{Z}_p$. Its transform is then supported on $\{l\}$, so $|u_\delta|$ is constant, and $\sum_{\delta \ne 0} \kappa_l(\delta) = p$ gives $h_T \le \frac{1}{2}$, with equality. □

So Proposition 22 is exact. Stray outputs at $T = [n]$ add the frequencies $2l \pm q$ ($q \in S$), and then $|M_{[n]}| \ge |B|$ at every sparse support computed ($p = 11, \dots, 41$ at $b = 2$). Hence every gain bound that is linear near $R = 0$ loses, at every sparse prime. The gain is in fact quadratic near the IF block. For three atoms within ε of the block, $|A| - 3^{1-s}Q^s = \frac{1-r}{8+r} |\beta_1 + \beta_2 + \overline{\beta_{12}}|^2/a + O(\varepsilon^3)$ (proved by second-order expansion; the slot phases cancel in that combination). The constants are 0.346 in terms of $\max_T |\beta_T|^2$ and 0.115 in terms of $\sum_T |\beta_T|^2$ at $b = 2$; the global supremum 0.275 is reached at moderate interference, on an explicit two-atom branch $G_2(\rho) = 3\rho - \rho^3/c^2 - 1$, $c = \frac{1}{2}(\frac{3}{2})^{s-1}$.

Numerical verdict ($n = 2$, every support feature of every prime $5 \le p \le 41$, $b = 2$ and 2.5). The prover wins in every case, using the sharp T-resolved gain (local search) as the network's payoff. The minimal ratio of protection to gain is

- 1.015 with partner and riding;
- 1.05 with stray outputs and joint Lorentz accounts;
- 1.69 with the quadratic test of Theorem 29.

The worst case is $p = 17$, $l = 5$, a two-neuron cheap feature with $|\beta_T| \approx 0.5$–$0.8\, a_l$. At $b = 2.5$ the minimal ratio is ≥ 2.35. Dedicated compensators never help once their siblings are counted, and

moving the allocation off $y^\star$ lowers the ratio by at most 7% (under 1% with the quadratic test). Two variants lower the ratio below 1. Making the own partner free gives 0.94 with single-character tests, still 1.66 with the quadratic test. Making every shared (F2) neuron free at any size gives 0.70, and 1.32 with the quadratic test. F2 relations $2l \equiv \pm j \pm k$ occur for 30 of the 38 features at $p \le 41$, so at these sizes they are the rule. What would make this a proof:

- a T-resolved sharp payment inequality on atom systems (single-feature and p-free; anything within $1.6\times$ of the computed bonus suffices for $p \le 41$ with the quadratic test);
- a p-uniform lower bound on the protection curves, i.e. a robust uncertainty principle on $\mathbb{Z}_p$. The slopes h_T vary irregularly with the arithmetic of (p, l), from 0.095 to 0.67.

Files: `mechanism_scripts/compensation/`.

A spectral form of the cost lemma. The paper proves $\sum_l |\beta_l|^r \ge \nu^{1-r}|\beta|^r$ for interference-free atoms by comparing $\det(I+tH)$, $H = \sum_l \beta_l m_l m_l^*$ (m_l the chain vectors), with $\prod_l (1+t\nu|\beta_l|)$. The comparison does not need interference-freeness. Since $|\det(I+tH)| = \prod_k |1+t\lambda_k|$ and $\int_0^\infty \log|1+te^{i\phi}|\, t^{-1-r}dt = \cos(r\phi)\,\frac{\pi}{r\sin\pi r}$ for $|\phi| \le \pi$ (rotate the contour), the same integration gives, for *any* atoms, any n and $0 < r < 1$,

$$\nu^r \sum_l |\beta_l|^r \;\ge\; \sum_{k=1}^{\nu} |\lambda_k|^r \cos(r \arg \lambda_k), \qquad \lambda_k \text{ the eigenvalues of } H.$$

H has constant diagonal $\beta = \sum_l \beta_l$, so $\sum_k \lambda_k = \nu\beta$, and its strictly lower entries are the sign-flipped coefficients β_T of the intervals T. The bound is an equality both for a single atom ($\lambda = (\nu\beta, 0, \dots, 0)$) and for the IF block ($\lambda = (\beta, \dots, \beta)$), the two extremes of the sharp single-feature constant. So at $n = 2$ the sharp payment $|\beta| \le 3^{1-s}Q^s + (1-3^{1-s})\max_T |\beta_T|$ would follow from the eigenvalue inequality $\sum_k \operatorname{Re}\lambda_k^r \ge 3\big(|\beta| - (1-3^{1-s})\max_T|\beta_T|\big)^r$ over 3×3 matrices with diagonal β, lower entries β_T and upper entries bounded by $\sum_l |\beta_l|$. The upper bound is needed: with free upper entries the spectrum can be anything with trace 3β. To second order the bound reads $|\beta| - \nu^{1-s}Q^s \lesssim (1-r)\,|\sum_T \beta_T U_T|/(\nu|\beta|)$, with U_T the upper entries. The bonus is thus driven by products of lower (sign-flipped) and upper entries, and both vanish at the IF block.

Quadratic tests: L^2 payment with factorised detection. A linear Riesz factor detects one character at a time, so it needs the pointwise payment (P), which is false at $n = 4$. Parseval (A_2) sees every character but loses $\sqrt{|L'|}$. A product of *quadratic* factors, one per cheap feature, does both. It detects the whole L^2 mass of each feature's interference, and it factorises across features under a dissociation hypothesis of order three. Put $f(\gamma) = \frac{1}{2}\big(\sqrt{\gamma^2+4} - \gamma\big)$. Then $f(0) = 1$, f is decreasing, and $f(\gamma) + f(-\gamma) \ge 2$.

Theorem 29 (quadratic Riesz products; class $\mathcal{C}$, every n)**.** *Let $p \ge 5$, $\theta \in \mathcal{C}$, $\delta \ne 0$ and $L' \subseteq F$. For $l \in L'$ let $W_l = \Phi_l - \overline{\beta_l} = \sum_{T \ne \emptyset} a_{l,T}\rho^{2ls_T}$, $\sigma_l^2 = \frac{1}{2}\sum_T |\beta_{l,T}|^2$,*

$$\tau_l = \sum_{(T,T')} a_{l,T}\, a_{l,T'}\, \overline{a_{l,T\cup T'}} \quad \textit{(ordered pairs of disjoint non-empty sets)}, \qquad \gamma_l(\delta) = \frac{3\,\operatorname{Re}\big(\tau_l\, c_l(\delta)\big)}{4\,\sigma_l^3\,|c_l(\delta)|}.$$

Suppose

(D_3) the only solution of $\sum_{l\in L'} \varepsilon_l\, l \equiv 0 \pmod p$ with integers $|\varepsilon_l| \le 3$, and $|\varepsilon_l| = 3$ for at most one l, is $\varepsilon = 0$;

(I_2) no $k \in F \setminus L'$ with $\Phi_k \not\equiv 0$ is congruent to some $\sum_{l\in L'} \varepsilon_l\, l$ with integers $|\varepsilon_l| \le 2$.

Then

$$h(\theta) \;\le\; \sum_{k\in F} \operatorname{Re}\big(\overline{\beta_k}\, c_k(\delta)\big) \;-\; \sum_{l\in L'} \sigma_l\, |c_l(\delta)|\, f\big(\gamma_l(\delta)\big)$$

(terms with $\sigma_l = 0$ are omitted).

Proof. Write $c_l = c_l(\delta)$ and $X_l = \operatorname{Re}(W_l c_l)$, so that $g(\cdot,\delta) = \sum_k \operatorname{Re}(\overline{\beta_k} c_k) + \sum_k X_k$. The monomials of W_l are $\rho^{\langle \chi, x\rangle}$ with $\chi = 2l\mathbf{1}_T$, and $\mathbb{E}\rho^{\langle\chi,x\rangle} = 0$ unless $\chi = 0$. For $p \ge 5$ the vectors $2l(\mathbf{1}_T + \mathbf{1}_{T'})$ and $2l(\mathbf{1}_T + \mathbf{1}_{T'} + \mathbf{1}_{T''})$ are non-zero (entries in $2l\{0,1,2,3\}$, not all 0), so $\mathbb{E}W_l^2 = \mathbb{E}W_l^3 = 0$. The vector $2l(\mathbf{1}_T + \mathbf{1}_{T'} - \mathbf{1}_{T''})$ vanishes iff $T \cap T' = \emptyset$ and $T'' = T \cup T'$, so $\mathbb{E}[W_l^2 \overline{W_l}] = \tau_l$. Hence $\mathbb{E}X_l = 0$, $V_l := \mathbb{E}X_l^2 = \frac{1}{2}|c_l|^2 \mathbb{E}|W_l|^2 = \sigma_l^2 |c_l|^2$ and $\mathbb{E}X_l^3 = \frac{3}{4}|c_l|^2 \operatorname{Re}(c_l\tau_l) = \gamma_l V_l^{3/2}$. For $t \in \mathbb{R}$ the density $w_l = (X_l - t)^2/(V_l + t^2)$ has $\mathbb{E}[w_l X_l] = (\mathbb{E}X_l^3 - 2tV_l)/(V_l + t^2)$. At $t = \frac{1}{2}\big(\gamma_l + \sqrt{\gamma_l^2 + 4}\big)\sqrt{V_l}$ this equals $-\sqrt{V_l}\, f(\gamma_l)$, which is its minimum.

Let $w = \prod_{l\in L'} w_l \ge 0$ and expand $w\, g(\cdot,\delta)$ into monomials. In slot t, a monomial of w_l has exponent $2l\, v_l$ with $|v_l| \le 2$, and a monomial of X_k has exponent $2k\, v$ with $|v| \le 1$. A monomial has non-zero mean only if every slot exponent vanishes. For the monomials of w alone, (D_3) forces all $v_l = 0$, so $\mathbb{E}w = \prod_l \mathbb{E}w_l = 1$. With a monomial of X_k, $k \notin L'$, a vanishing exponent with $v \ne 0$ would contradict (I_2), so $\mathbb{E}[wX_k] = 0$. With a monomial of X_k, $k \in L'$, the exponent is $\sum_{l\ne k} 2l\, v_l + 2k(v_k + v)$ with $|v_k + v| \le 3$. Then (D_3) forces $v_l = 0$ for $l \ne k$, so $\mathbb{E}[wX_k] = \mathbb{E}[w_k X_k]$. Therefore $h \le \mathbb{E}[w\, g(\cdot,\delta)] = \sum_k \operatorname{Re}(\overline{\beta_k} c_k) - \sum_{l\in L'} \sqrt{V_l}\, f(\gamma_l)$. □

Corollary 30 (class $\mathcal{C}$: any number of dissociated cheap features)**.** *Let $p \ge 5$ and $2 \le b \le \nu$. Define*

$$k^\sharp(n,s) := \sup \frac{\operatorname{Re} A - \nu^{1-s} Q^s}{\sigma\, f(\gamma_{\max})}, \qquad \gamma_{\max} = \frac{3|\tau|}{4\sigma^3},$$

over atom systems with $\sigma > 0$, with A, Q, β_T as in Theorem 14 of the companion note, $\sigma^2 = \frac{1}{2}\sum_T |\beta_T|^2$ and $\tau = \sum_{(T,T')} \beta_T \beta_{T'} \overline{\beta_{T\cup T'}}$. If $k^\sharp(n,s) < 1$, then no network of $\mathcal{C}$ whose cheap set L' satisfies (D_3) and (I_2) beats the interference-free value, and equality forces $L' = \emptyset$. For $n = 2$ and $2 \le b \le 3$,

$$k^\sharp(2,s) \;\le\; \frac{1}{\sqrt{7} - 1} \;=\; 0.608.$$

So at $n = 2$, among networks of $\mathcal{C}$ with (D_3)- and (I_2)-dissociated cheap features (any number of them), every maximiser is interference-free and uses $K^*(s,p)$ features.

Proof. Let $\delta^\star$ minimise $\sum_l a_l \kappa_l(\delta)$. Average Theorem 29 at $\pm\delta^\star$. Then $\frac{1}{2}\sum_\pm \operatorname{Re}(\overline{\beta_k} c_k(\pm\delta^\star)) = \operatorname{Re}\beta_k\, \kappa_k(\delta^\star)$ and $|c_l(-\delta)| = |c_l(\delta)|$. Since $|\operatorname{Re}(\tau_l c)/|c|| \le |\tau_l|$, $\gamma_l(\pm\delta^\star) \le \gamma_{\max,l}$ and $f(\gamma_l(\pm\delta^\star)) \ge f(\gamma_{\max,l})$. For $l \notin L'$, $\operatorname{Re}\beta_l \le a_l$. For $l \in L'$, by definition $\operatorname{Re}\beta_l - a_l \le k^\sharp \sigma_l f(\gamma_{\max,l})$, and τ_l has the modulus of the τ of l's atoms. Hence

$$h \;\le\; \sum_l a_l \kappa_l(\delta^\star) + \sum_{l\in L'} \sigma_l\, f(\gamma_{\max,l})\big(k^\sharp \kappa_l(\delta^\star) - |c_l(\delta^\star)|\big),$$

and $\kappa \le \sqrt{2\kappa} = |c|$ because $\kappa \le 2$. Each bracket is negative, and $\sigma_l > 0$ for $l \in L'$ (the paper's Lemma 7). Conclude as in Theorem 17. For $n = 2$: $|\tau| = 2|\beta_1\beta_2\beta_{12}| \le 2(2\sigma^2/3)^{3/2}$ by AM–GM, so $\gamma_{\max} \le \sqrt{2/3}$ and $f(\gamma_{\max}) \ge (\sqrt{14} - \sqrt{2})/(2\sqrt{3}) = 0.672$. Part I's Theorem A gives $\operatorname{Re} A - 3^{1-s}Q^s \le \sigma/\sqrt{6}$. The ratio is $\sqrt{2}/(\sqrt{14} - \sqrt{2}) = 1/(\sqrt{7} - 1)$. □

Three remarks place this in the architecture.

(a) *A uniform, proved surplus at $n = 2$.* Per feature and direction, detection exceeds payment by at least $\sigma_l f\,|c_l|\big(1 - k^\sharp \kappa_l/|c_l|\big) \geq (1 - 0.608)\,\sigma_l f\,|c_l|$, i.e. by 39% of the detection, whatever κ_l. The linear Riesz test has a surplus that vanishes as $\kappa_l \to 2$ with Theorem A's constant, and is 18% with the sharp one (Theorem 34). The quadratic test gives a uniform surplus from Theorem A alone. Local search gives $k^\sharp(2, \frac{3}{2}) \geq 0.514$, so the true surplus is at most about 49%.

(b) *No $\sqrt{|L'|}$ loss, for every n.* The test is the L^2 route of Theorem 17 with the Cauchy–Schwarz step replaced by a product, so it detects $\sum_l \sigma_l |c_l| f$, linearly in the features. The price is order-three dissociation (D_3), and a single-feature constant $k^\sharp(n, s) < 1$ in place of $c_1 \leq \frac{1}{2}$. The constant $k^\sharp$ is defined on atoms and does not depend on p. The crude bound $k^\sharp \leq k^*/f(\gamma_{\max}(n))$ uses $\gamma_{\max}(n) = \sqrt{2/3},\ \sqrt{2},\ 1.96,\ 2.49$ for $n = 2, \dots, 5$ (cubic-form maximum on the sphere, computed). With $k^*(3, 2) \approx 0.54$ this crude bound is already about 1.04 at $n = 3$, so for $n \geq 3$ the joint supremum $k^\sharp$ is what has to be computed.

(c) *General networks.* For an arbitrary network take X_l to be the part of $g(\cdot, \delta)$ on feature l's characters $\pm\chi_{l,T}$: own modes, partner and foreign modes together. The same computation gives detection $\sqrt{V_l} f(\gamma_l)$ with $V_l = 2\sum_T |\hat{g}_\delta(\chi_{l,T})|^2$, and $\gamma_l \leq \gamma_{\max}(n)$ for every coefficient vector. Two defects appear. Compensation lowers V_l, as in Lemma 25 but now in L^2 over T. Modes of g at the pair sums $\pm\chi_{l,T} \pm \chi_{l,T'}$ and at order-three relations among the cheap frequencies are of type (R). So (O1) and (O2) keep their form, and they now have to fit inside the surplus of (a).

(d) *Hermitian squares.* The proof never uses the special form $(X_l - t)^2$. Any factor $w_l = |P_l|^2/\mathbb{E}|P_l|^2$ with P_l in the span of 1 and the characters $\pm\chi_{l,T}$ has its monomials at differences of two such characters, so it factorises under the same (D_3), (I_2). The best such factor detects $-\lambda_{\min}(A_l(\delta))$, where $A_l(\delta)$ is the Hermitian matrix $\big(\hat{X}_l(v - u)\big)_{u,v}$ indexed by $\{0\} \cup \{\pm\chi_{l,T}\}$, of size $2^{n+1} - 1$. It contains the quadratic factor ($P = X_l - t$) and the linear Riesz factor ($1 - \cos = \frac{1}{2}|1 - e^{i(\cdot)}|^2$), so it dominates both. After averaging $\pm\delta$ (so $\kappa_l = 2\cos^2\alpha$, $|c_l| = 2\cos\alpha$ with $\alpha = |\arg c_l(\delta)|$), Corollary 30 holds with $k^\sharp$ replaced by the direction-resolved constant

$$k^\flat(n, s) = \sup_{\text{atoms},\ 0 \leq \alpha < \pi/2} \frac{(\operatorname{Re} A - \nu^{1-s} Q^s)\cos\alpha}{\frac{1}{2}\big(D(\alpha) + D(-\alpha)\big)},$$

with $D(\pm\alpha)$ the best Hermitian-square detection for $|c| = 1$ and $\arg c = \pm\alpha$. Local search at $b = 2$ gives $k^\flat \geq 0.423$ at $n = 2$ (this is $c_1(2, \frac{3}{2}) = 1 - 3^{-1/2}$, attained by a single neuron) $k^\flat \geq 0.535$ at $n = 3$, 0.559 at $n = 4$ and 0.570 at $n = 5$ (two or three atoms, partly cancelled; at $n = 4$ two slots form an interference-free block and the interference sits on $T \subseteq \{3, 4\}$). The quadratic factor alone, direction-resolved, gives 0.514, 0.762, 1.006 for $n = 2, 3, 4$.

The relations case at $n = 2$: what is proved. The *relations case* is class $\mathcal{C}$ at $n = 2$ with every feature's interference on $T = \{1, 2\}$. This is the case in which the network can force additive relations among cheap features. Then $\Phi_l = \overline{\beta_l} + W_l \rho^{2l(x_1+x_2)}$. Put $w = 2(x_1+x_2)$, $v_l = |W_l| = |\beta_{l,12}|$, $\varphi(w) = \sum_l v_l \cos(2\pi l w/p + \arg W_l)$ and $D_\delta(w) = \varphi(w) - \varphi(w-\delta)$, so that $g(x, \delta) = \mathbb{E}g(\cdot, \delta) + D_\delta(w)$.

Lemma 31 (oscillation)**.** *In the relations case, $h \leq \sum_l \kappa_l(\delta) \operatorname{Re}\beta_l - \frac{1}{2}\operatorname{osc}_w D_\delta(w)$ for every $\delta \neq 0$, where* osc = max − min.

Proof. Use separate minimisers for $\pm\delta$, and note $D_{-\delta}(w) = \varphi(w) - \varphi(w + \delta) = -D_\delta(w + \delta)$, so $\min D_{-\delta} = -\max D_\delta$. □

The quantity is two-sided, and $\mathbb{E}_w D_\delta^2 = \sum_l v_l^2 \kappa_l(\delta)$ exactly: the frequencies $l \in F$ are distinct and $l \not\equiv -l'$, so no relation enters. Since $\operatorname{osc} D \geq 2\sqrt{\mathbb{E}D^2}$ (Bhatia–Davis), the proof of Theorem 17 with one set T instead of $2^n - 1$ gives:

Corollary 32 (Parseval closure)**.** *Let c be a valid single-T payment constant ($\operatorname{Re}\beta_l \leq a_l + c\, v_l$; Theorem A gives $c = 1/(2\sqrt{3})$). In the relations case, if $c^2 \sum_{l \in L'} \kappa_l(\delta^\star) < 1$ then $\gamma(\theta) < \nu^{1-s} B_p \Gamma_s$ unless $L' = \emptyset$. Since $\sum_{l \in F} \kappa_l(\delta) = p/2$:*

(a) the relations case is closed for every prime $p \leq 23$, every cheap set, every $b \in [2, 3]$ and any width;

(b) at every prime it is closed whenever the cheap set carries $\sum_{L'} \kappa_l(\delta^\star) < 12$, in particular whenever $|L'| \leq 6$;

(c) with all sets T allowed, Theorem 17 (threshold $4 \geq p/2$) closes all of class $\mathcal{C}$ at $n = 2$ for $p \in \{5, 7\}$: every maximiser within $\mathcal{C}$ is interference-free and uses $K^*(s, p)$ features.

Theorem 33 (sharp single-T payment, P_{12})**.** *Let $\beta_i \in \mathbb{C}$ and $\xi_i, \eta_i \in \mathbb{T}$ with $B = \sum_i \beta_i \xi_i = 0$ and $C = \sum_i \beta_i \eta_i = 0$. Put $A = \sum_i \beta_i$, $D = \sum_i \beta_i \xi_i \eta_i$, $1 \leq s \leq \frac{3}{2}$, $r = 1/s$, $\theta = 3^{1-s}$ and $c_{12} = 1 - (\frac{2}{3})^{s-1}$. Then $|A| - c_{12}|D| \leq \theta(\sum_i |\beta_i|^r)^s$, with equality for two aligned atoms.*

Proof. Write $\beta_i = \rho_i p_i$ ($\rho_i > 0$, $p_i \in \mathbb{T}$) and $L = \sum \rho_i$. *Twist* the chain vectors by the atom's own phase: $u_i = (1, x_i, y_i)$ with $x_i = p_i \xi_i$, $y_i = p_i^2 \xi_i \eta_i$, and $H = \sum_i \rho_i u_i u_i^* \succeq 0$. Its diagonal is L, and its entries below the diagonal are $\sum \rho_i x_i = B = 0$, $\sum \rho_i y_i \bar{x}_i = C = 0$ and $D' := \sum \rho_i y_i$. So $H = \left(\begin{smallmatrix} L & 0 & \bar{D}' \\ 0 & L & 0 \\ D' & 0 & L \end{smallmatrix}\right)$. Put $\delta = |D'|/L$, $\omega = D'/|D'|$ and $X_i = \operatorname{Re}(\bar{\omega} y_i)$.

Cost. Leverage scores are at most one: $\rho_i w_i \leq 1$ with $w_i = u_i^* H^+ u_i = (3 - \delta^2 - 2\delta X_i)/(L(1-\delta^2))$. So $\rho_i^r \geq \rho_i w_i^{1-r}$. The map $X \mapsto w^{1-r}$ is concave on $[-1, 1]$, and $\sum \rho_i X_i = L\delta$. Comparing with the chord gives $Q \geq L^r \Phi(\delta)$, $\Phi(\delta) = \frac{1}{2}[(1+\delta)^r (3+\delta)^{1-r} + (1-\delta)^r (3-\delta)^{1-r}]$; the case $\delta = 1$ is a limit.

Gain. We have $D = \sum \rho_i \bar{p}_i y_i$. With $P_i = \bar{\alpha} p_i$ ($\alpha = A/|A|$) and $Y_i = \bar{\omega} y_i$, Cauchy–Schwarz gives, for every $C' \geq 0$,

$$|A| - C'|D| \leq \sum_i \rho_i \operatorname{Re}\left(\bar{P}_i (1 - C' Y_i)\right) \leq \sum_i \rho_i |1 - C' Y_i| \leq L\sqrt{1 + C'^2 - 2C'\delta}.$$

One variable. It remains to show $3^{1-s}\Phi(\delta)^s \geq \sqrt{1 + C'^2 - 2C'\delta}$ for $C' = \min(\delta, c_{12})$.

- For $\delta \leq c_{12}$ this is $\varphi := 3^{r-1}\Phi \geq (1 - \delta^2)^{r/2}$. It follows from $\varphi = (1 - \delta^2)^{r/2}(1 - \frac{\delta^2}{9})^{\frac{1-r}{2}} \cosh(r \operatorname{artanh}\delta + (1 - r)\operatorname{artanh}\frac{\delta}{3})$.
- For $\delta \geq c_{12}$, Jensen gives $\varphi \geq e^{(1-r)\kappa(\delta)}$ with $\kappa = \frac{1+\delta}{2}\log\frac{3+\delta}{3(1+\delta)} + \frac{1-\delta}{2}\log\frac{3-\delta}{3(1-\delta)}$. Here e^κ is concave ($\kappa'' + \kappa'^2 \leq -\frac{5}{16}$) and $\kappa(1) = \log\frac{2}{3}$. Then $x = (\frac{2}{3})^{s-1}$ enters as $x^{2\kappa/\log(2/3)}$, which is concave in x, so the inequality reduces to its endpoints in x. These are elementary.

Chaining the three steps: $|A| - c_{12}|D| \leq |A| - C'|D| \leq L\sqrt{1 + C'^2 - 2C'\delta} \leq \theta(L^r\Phi)^s \leq \theta Q^s$. Full details, and checks of every step on $2 \cdot 10^5$ random configurations, are in `mechanism_scripts/payment/`. □

Theorem 34 (sharp pointwise payment at $n = 2$; computer-assisted)**.** *For all atoms $\beta_i \in \mathbb{C}$, $\xi_i, \eta_i \in \mathbb{T}$ (no constraint) and $1 \leq s \leq \frac{3}{2}$:*

$$|A| \;\leq\; 3^{1-s} Q^s + \left(1 - 3^{1-s}\right) \max(|B|, |C|, |D|),$$

with equality for a single atom. So $c_1(2, s) = 1 - 3^{1-s}$ (0.4226 at $b = 2$).

Proof outline. Use the same twisted matrix $H = L\hat{H}$. Now $\hat{H}$ is a correlation matrix with entries B/L, C/L, D'/L below the diagonal and eigenvalues $\nu_k \geq 0$, $\sum \nu_k = 3$.

- The paper's determinant comparison applies to H (real non-negative weights), so $\theta Q^s \geq L(\frac{1}{3}\sum \nu_k^r)^s$. Jensen with weights $\nu_k/3$ turns this into $\geq L\theta^\lambda$, $\lambda = \sum \nu_k \log \nu_k/(3\log 3)$.
- The spectrum solves $\mu^3 - P\mu - q = 0$ with $P = 2b^2 + y^2$, $b = \sqrt{(|B|^2+|C|^2)/2}/L$, $y = |D'|/L$ and $q \geq \max(-2b^2 y, P-1)$. By divided differences, λ increases in P and decreases in q, which gives $\lambda \leq \tilde{\lambda}(b,y)$, increasing in both.
- The Cauchy–Schwarz step above (which does not use $B = C = 0$) gives $|A| - c_1 M \leq L[\sqrt{1 + C'^2 - 2C'y} - c_1\lambda' b]$ for admissible (λ', C').
- Both sides are concave in $x = 3^{1-s}$, which removes s. What remains is one inequality in $(b,y) \in [0,1]^2$. Near the two equality corners (IF block, single atom) it is proved analytically. Elsewhere it is certified by interval arithmetic: every box passes with minimal margin $2.47 \cdot 10^{-4}$, and we re-ran the certificate.

□

Consequences. With $c_{12} = 0.1835$ ($b = 2$) instead of $1/(2\sqrt{3}) = 0.289$, the threshold of Corollary 32 rises to ≥ 29.7. So the relations case is **closed for every prime $p \leq 59$, for all networks**, and at every prime whenever the cheap set carries $\sum \kappa_l < 29.7$, in particular whenever $|L'| \leq 14$. Theorem 34 turns the conditional statements made above with the sharp constant into theorems. The linear Riesz test now has a proved uniform surplus of $(\varepsilon - 2c_1)\frac{\kappa_l}{2}|\beta_{l,T}|$ in every direction, 18% of the payment at $b = 2$. And with partner and riding compensation at no cost, a single cheap feature is excluded at $p = 5, 7, 13$ (the free-hiding values $V(c_1) > 0$). The same chain also gives a global quadratic bound, bonus $\leq K \sum_T |\beta_T|^2/|A|$ with $K \approx 0.282$ at $b = 2$ (numerical maximisation of a proved bound; the true supremum is about 0.275).

A buffer near the interference-free networks, at every prime. The sharp constants above are attained at a single atom or at two aligned atoms, i.e. far from the IF blocks. Near an IF block a feature's discount is *quadratic* in its interference, while the slack that the interference forces is *linear* (Lemma 31, Bhatia–Davis). So small departures from the predicted network lose by a whole order, and no sharp constant and no arithmetic are needed there.

Proposition 35 (quadratic payments, $n = 2$)**.** *Let $1 \leq s \leq \frac{3}{2}$, $r = 1/s$, $\theta = 3^{1-s}$, let $\beta_i \in \mathbb{C}$, $\xi_i, \eta_i \in \mathbb{T}$ be atoms with A, B, C, D, Q as in Theorem 34, and $L = \sum_i |\beta_i|$.*

(i) $|A| - \theta Q^s \leq K_2\,(|B|^2 + |C|^2 + |D|^2)/L$ *with* $K_2 = 2(s-1)/(7-4s)$ *(= 1 at $s = \frac{3}{2}$).*

(ii) If $B = C = 0$, then $|A| - \theta Q^s \leq K_q\,|D|^2/L$ *with* $K_q = 5/(5+4r)$ *(≤ 0.652; computer-assisted).*

Adversarial search (600 starts, up to 8 atoms) gives the suprema 0.2805 *in (i) and* exactly $c_{12} = 1 - (\frac{2}{3})^{s-1}$ *in (ii), at $s = 1.1, 1.25, \frac{3}{2}$, attained by two aligned atoms. So (ii) presumably holds with c_{12} in place of K_q: $|A| - \theta Q^s \leq c_{12}|D| \cdot |D|/L$, a common strengthening of P_{12} and of (ii).*

Proof. (i) Take the twisted matrix $H = \sum_i \rho_i u_i u_i^*$ of Theorem 33 and $\hat{H} = H/L$. It has unit diagonal, entries B/L, C/L, D'/L below the diagonal, and eigenvalues $\nu_k \geq 0$ with $\sum \nu_k = 3$. The

spectral cost lemma with real weights gives $\theta Q^s \ge L(\frac{1}{3}\sum_k \nu_k^r)^s$. Jensen with weights $\nu_k/3$, then $e^{-x} \ge 1-x$ and $\nu\log\nu \le (\nu-1)+(\nu-1)^2$, give

$$\theta Q^s \;\ge\; L\exp\Big(-(s-1)\sum_k \tfrac{\nu_k}{3}\log\nu_k\Big) \;\ge\; L - \tfrac{s-1}{3}L\sum_k(\nu_k-1)^2 \;=\; L - \tfrac{2(s-1)}{3L}\big(|B|^2+|C|^2+|D'|^2\big).$$

Put $\alpha = A/|A|$ and $P_i = \bar\alpha p_i$. Then $L-|A| = \frac{1}{2}\sum_i \rho_i|1-P_i|^2$, and $D' = \alpha^2\sum \rho_i P_i^2\xi_i\eta_i$, $D = \alpha\sum\rho_i P_i\xi_i\eta_i$ give $|D'| \le |D| + \sum_i \rho_i|P_i-1| \le |D| + \sqrt{2L(L-|A|)}$. With $\eta = 4(s-1)/(7-4s)$, $|D'|^2 \le (1+\eta)|D|^2 + (1+\frac{1}{\eta})2L(L-|A|)$, and $\frac{4(s-1)}{3}(1+\frac{1}{\eta}) = 1$, so the terms in $L-|A|$ cancel: $|A| - \theta Q^s \le \frac{2(s-1)}{3L}(|B|^2+|C|^2+(1+\eta)|D|^2) \le K_2(|B|^2+|C|^2+|D|^2)/L$, because $1+\eta = 3/(7-4s)$.

(ii) The proof of Theorem 33 gives $Q \ge L^r\Phi(\delta)$ and $|A| - C'|D| \le L\sqrt{1+C'^2-2C'\delta}$ for every $C' \ge 0$, with $\delta = |D'|/L$. Take $C' = \delta$ and $\varphi = 3^{r-1}\Phi$. Then

$$|A| - \theta Q^s \;\le\; \delta|D| - L\,g(\delta), \qquad g(\delta) = \varphi(\delta)^s - \sqrt{1-\delta^2}.$$

The coefficient of δ^2 in g at 0 is $\kappa_0 = \frac{1}{2} - \frac{2}{9}(1-r) = \frac{5+4r}{18}$. We certify $g(\delta) \ge \kappa_1\delta^2$ on $[0,1]$ with $\kappa_1 = 0.9\kappa_0$, for all $s \in [1, \frac{3}{2}]$. Then $|A| - \theta Q^s \le \max_\delta(\delta|D| - \kappa_1 L\delta^2) = |D|^2/(4\kappa_1 L) = K_q D^2/L$. The certificate uses interval arithmetic in three ranges:

- $\delta \le 0.21$: Bernoulli's $\varphi^s \ge 1 - s(1-\varphi)$, the identity $1-\sqrt{1-\delta^2} = \delta^2/(1+\sqrt{1-\delta^2})$, and $1-\varphi(\delta) = -3^{r-1}f''(\xi)\delta^2/2$ for some $|\xi| \le \delta$, where $f(x) = (1+x)^r(3+x)^{1-r}$;
- $0.21 \le \delta \le 0.5$: evaluation at the centres of 200 s-boxes, plus a mean-value bound in s;
- $\delta \ge 0.5$: direct evaluation on $10^3 \times 200$ boxes.

All 237,400 boxes pass. The smallest certified margins are $1.4\cdot10^{-4}$ for $g/\delta^2 - \kappa_1$ on $\delta \le 0.21$ and $5.3\cdot10^{-4}$ for $g - \kappa_1\delta^2$ beyond (`mechanism_scripts/nearif/`). □

Theorem 36 (near-IF rigidity at $n=2$; every prime, any number of cheap features)**.** *Let $n=2$, $2 \le b \le 3$, p any odd prime and θ any network of any width. Let $\delta^\star$ minimise $\sum_l a_l\kappa_l(\delta)$, let L' be the cheap set, and let $L_l = \sum_i|\beta_{il}|$ be the atom mass of feature l.*

(a) Relations case. *Put $v_l = |\beta_{l,12}|$ and $t_l = v_l/L_l \in [0,1]$. If*

$$\sum_{l\in L'} \min(c_{12}, K_q t_l)^2\,\kappa_l(\delta^\star) \;<\; 1,$$

then $\gamma(\theta) \le \nu^{1-s}B_p\Gamma_s(p)$, and equality forces θ to be interference-free. In particular this holds whenever $\big(\sum_l v_l^2\kappa_l(\delta^\star)\big)^{1/2} < \min_{l\in L'} L_l/K_q$.

(b) Class $\mathcal{C}$. *Put $E_l = \sum_{T\neq\emptyset}|\beta_{l,T}|^2$. If $\big(\sum_l E_l\kappa_l(\delta^\star)\big)^{1/2} < \min_{l\in L'} L_l/(\sqrt{3}K_2)$, the same conclusion holds.*

Proof. (a) Lemma 31 at $\delta^\star$ and Bhatia–Davis give $h \le \sum_l \mathrm{Re}\,\beta_l\kappa_l(\delta^\star) - X$ with $X^2 = \mathbb{E}D^2_{\delta^\star} = \sum_l v_l^2\kappa_l(\delta^\star)$. For $l \in L'$, Theorem 33 and Proposition 35(ii) give $\mathrm{Re}\,\beta_l - a_l \le \min(c_{12}v_l, K_q v_l^2/L_l) = \min(c_{12}, K_q t_l)\,v_l$. By Cauchy–Schwarz,

$$h \;\le\; \sum_l a_l\kappa_l(\delta^\star) + X\Big(\big(\textstyle\sum_{l\in L'}\min(c_{12},K_q t_l)^2\kappa_l(\delta^\star)\big)^{1/2} - 1\Big),$$

and we conclude as in Theorem 17: $X=0$ means no interference. For the special case, $\min(c_{12},K_q t_l)v_l \le K_q v_l^2/L_l$, so the bonus is at most $K_q X^2/\min_{L'} L_l < X$. (b) The proof of Theorem 17 gives $h \le \sum_l a_l\kappa_l(\delta^\star) + \sum_{l\in L'}(\mathrm{Re}\,\beta_l - a_l)\kappa_l(\delta^\star) - \bar m$ with $\sum_l E_l\kappa_l(\delta^\star) \le 3\bar m^2$. Proposition 35(i) bounds the bonus by $K_2\sum_{L'} E_l\kappa_l/L_l \le K_2X^2/\min_{L'}L_l$, where now $X^2 = \sum_l E_l\kappa_l(\delta^\star)$, and $\bar m \ge X/\sqrt{3}$. □

The conditions do not depend on p, on the width or on $|L'|$. Corollary 32 with $c = c_{12}$ is implied by (a), since $\min(c_{12}, K_q t_l) \le c_{12}$. What is left of the relations case is therefore networks with $\sum_{L'} \min(c_{12}, K_q t_l)^2 \kappa_l \ge 1$. This needs $\sum_{L'} \kappa_l \ge 1/c_{12}^2 = 29.7$, i.e. some thirty cheap features, most of them with interference comparable to their atom mass. If (ii) holds with c_{12}, the condition becomes $\sum_{L'} t_l^2 \kappa_l \ge 29.7$. In terms of the network, the only way left to beat the prediction is to rebuild dozens of circuits at once from fewer neurons than the IF block uses, with interference phases arranged so that the profile φ is flat. Any network near the predicted one loses, at every prime.

A route for what is left (heuristic). Flatness at that level needs additive structure. A real Sidon constant below c_{12} needs at least fifteen frequencies (Parseval), and in practice many more with many short relations: Rudin–Shapiro-type polynomials have sup norm about $\sqrt{2}$ times the ℓ^2 norm. By the Laplace identity below, relation-rich sets cover poorly, so such a block is nearly blind on a large Bohr-type set of offsets. The margin is a minimum over δ. So the binding offsets either move into that blind set, where the block contributes neither coverage nor discount (its weights $\kappa_l(\delta)$ are small there), or the rest of the support must over-cover it. Over-covering a set of p/N offsets beats $\Gamma_s(p)$ by a factor of about $1 + (s-1)\ln N/\ln p$ at most. That caps the block's share of the budget at $f \lesssim \frac{s-1}{s} \ln N/\ln p$. With the concave cost, a block that thin gains $O(f^s)$ and costs $\Omega(f)$. Each step has a precise form: an inverse theorem for flat polynomials via the tilt, a covering bound for Bohr sets, and the dilution estimate. Together they would close the relations case at every prime. None of them is proved here.

The one-sided constant is the wrong object. For the full set F the one-sided constant $\min_{b,\theta}(-\min_w \sum b_l \cos(2\pi l w/p - \theta_l))/\sum b_l$ is $1/(p-1)$, attained at zero phases. There $\sum_{l\in F} \cos(2\pi l w/p)$ is a spike, so perfect coverage gives a *small* one-sided constant. On the programme supports it is $0.25, 0.17, 0.60, 0.34, 0.48, 0.52, 0.35, 0.46, 0.44$ for $p = 5, \dots, 31$, below $\frac{1}{2}$ at most primes and below 0.18 at $p = 7$, where the relations case is nevertheless closed. So (O1) as first stated, with a one-sided constant and a common evaluation point, is false. The correct object is the oscillation of the first difference (Lemma 31), and a spike φ has $\operatorname{osc} D_\delta \approx 2 \max \varphi$.

Coverage controls relations through a Laplace transform. For $S \subseteq \mathbb{Z}_p$ and $x \in \mathbb{R}^S_{\ge 0}$,

$$\sum_{k\in\mathbb{Z}^S:\ \sum_l k_l l \equiv 0} \prod_l I_{|k_l|}(x_l) \;=\; \frac{1}{p} \sum_{\delta\in\mathbb{Z}_p} e^{\sum_l x_l \cos(2\pi l\delta/p)} \;=\; e^{\sum_l x_l}\, \frac{1}{p} \sum_{\delta} e^{-\sum_l x_l \kappa_l(\delta)},$$

by $e^{x\cos\vartheta} = \sum_k I_k(x) e^{ik\vartheta}$ and orthogonality. The total Bessel weight of all additive relations is the Laplace transform of the coverage profile. Testing $P(w) = \sum_l b_l \cos(2\pi l w/p + \psi_l)$ with $e^{\lambda P}$ gives, uniformly in the phases, $\max P, -\min P \ge \frac{1}{\lambda} \log\big(2\prod_l I_0(\lambda b_l) - Z\big)$, with Z the left side above at $x = \lambda b$. In particular, write $E = \sum_l (x_l - \log I_0(x_l))$. If $E \le \ln \frac{p}{2}$ and $\min_{\delta\ne 0} \sum_l x_l \kappa_l(\delta) \ge E + \ln 2$, then $\max P, -\min P \ge \frac{1}{\lambda} \sum_l \log I_0(\lambda b_l)$. This is the precise sense in which coverage implies a Sidon-type bound: through an exponential tilt, not a Riesz product. A linear Riesz product cannot be controlled by minimum coverage at all. At the programme scale $K \approx 1.5 \ln p$ the tilt certifies 0.40 of $\sum_l b_l$. That is enough with P_{12} (Theorem 33; needed 0.367) but not with Theorem A (needed 0.577); with the computed supports it also needs the averaged rather than the minimum coverage.

Product tests are the wrong tool here. In the relations case the quadratic factor of Theorem 29 is exactly $1 - \sqrt{2}\varepsilon\cos u + \frac{\varepsilon}{2}\cos 2u$. It has off-zero weight 1.91ε against ε for the linear factor, and with worst-case phases its relation leakage exceeds its surplus on most programme supports for $p \le 97$. Parseval needs no relation accounting. Beyond Corollary 32 only cheap sets spread over most of F remain. For those, a phase-free sufficient condition obtained by water-filling the Parseval bound has supremum ratio 0.80–0.84 to the IF value for $p = 29, \dots, 61$ (local search; maximisers spread over 11–16 frequencies). So spread cheap sets lose to the coverage loss of a spread allocation, by at

least 15%. Crude scaling puts the failure of every L^2 argument at $\ln p \gtrsim 11$, where an extreme-value bound such as the tilt is needed. Files: `mechanism_scripts/relations2/`.

Only efficient allocations matter, and then the relations case is a flat-polynomial question. Let $\sigma_\delta(S)$ be the infimum, over $b \geq 0$ supported on S and all phases, of $\frac{1}{2}\operatorname{osc}_w\big(\sum_l b_l \cos(2\pi l w/p + \psi_l)\big)/\sum_l b_l s_l(\delta)$, with $s_l(\delta) = |\sin(\pi l\delta/p)|$. Let $e(a) = \min_\delta (Ka)_\delta/(\Gamma_s \|a\|_{1/s}^s) \leq 1$ be the efficiency of the allocation.

Proposition 37 (efficiency–Sidon reduction; every p)**.** *In the relations case, the allocation a cannot beat the IF value with any interference $v \leq \tau a$ ($\tau = 3^{s-1}$) and any phases if $\sigma_{\delta_0}(\operatorname{spt} a) \geq c - (1/e(a) - 1)/\tau$ at a minimiser δ_0 of $(Ka)_\delta$. Here c is the single-T payment constant. In particular every allocation with $e(a) \leq 1/(1 + c\tau)$ is closed, with no arithmetic input: $e \leq \frac{2}{3}$ under Theorem A, $e \leq 0.759$ with P_{12} ($b = 2$).*

Proof. At δ_0, Lemma 31 and $\sum_l b_l s_l = \sum_l v_l \kappa_l$ (with $b_l = 2v_l s_l$) give $h \leq (Ka)_{\delta_0} + (c - \sigma)(Kv)_{\delta_0} \leq (Ka)_{\delta_0}\big(1 + (c - \sigma)^+\tau\big) \leq (Ka)_{\delta_0}/e(a) = \Gamma_s\|a\|$. □

So the relations case asks for a flat-polynomial bound only on the supports of near-optimal allocations; Parseval gives $\sigma_\delta \geq (\sum_S \kappa_l(\delta))^{-1/2}$. The tilt can be improved to a tilted mean. If $M(\lambda) = \mathbb{E}_w e^{\lambda P}$ is the relation sum, then M'/M is a tilted mean of P, so it is at most $\max P$. Every relation term is increasing in λ, so $\max P, -\min P \geq (2\Pi' - Z')/Z$, with $\Pi = \prod_l I_0(\lambda b_l)$ and Z the Laplace transform above. The relation excess has a probabilistic form: $Z/\Pi - 1 = \Pr(\kappa \in \mathrm{Rel} \setminus 0)/\Pr(\kappa = 0)$ for independent Skellam variables, which is e^E/p plus the excess over independent phases.

Two limits. First, *any* argument that bounds relation terms by their moduli (tilts, positive-definite product tests) certifies at most about $(\ln p/K)\sum_l b_l$ once $K \gg \ln p$, because $Z \geq e^{\lambda \sum b}/p$. Under Theorem A's constant, closing the relations case therefore needs phase cancellation, i.e. a genuine flat-polynomial bound for sets of size $O(\log p)$. Second, the exact moments $\sum_\delta \kappa_l(\delta)$ and $\sum_\delta \kappa_l \kappa_k$ are the same for every support, and the progression $\{1, \dots, K\}$ has averaged exponential coverage far above e^{-E}. So a relation-mass hypothesis (RM: the excess over independent phases is at most e^E/p) has to come from near-optimality, not from moments. Numerically:

- at the programme vertices for all 33 primes up to 1009, $\sigma \geq 0.367 > c_*$ (Parseval alone), so cheap sets inside the support are closed;
- (RM) holds on programme supports for $23 \leq p \leq 10007$, with excess ratio ≤ 0.97; random sets of the same size reach 1.38;
- under Theorem A the water-filling bound holds up to $p = 10007$, and is marginal at $p = 30011$ (1.02–1.03, within the uncertainty of Γ_s), so the L^2 edge is near $p \approx 10^4$–$3 \cdot 10^4$;
- with P_{12} every case tested closes, and in a large-deviation model ($K = \kappa \ln p$, $p \to \infty$) P_{12} with the tilt and (RM) closes every κ with margin at least 1.10, while Theorem A with the tilt fails for $4 \lesssim \kappa \lesssim 17$;
- an adversarial search over amplitudes and phases finds $\sigma \approx 0.66$–0.69 on near-optimal supports where 0.25 is needed. So the inequality has a lot of room, and the proofs are what is lossy.

Since P_{12} is now proved (Theorem 33), the relations case at every prime reduces to the relation-mass hypothesis (RM) for efficient allocations. Files: `mechanism_scripts/relations3/`.

Local optimality cannot decide the count. At a local maximiser, Fritz John and Euler give a probability q on the active set with $\sum q\nabla g = \lambda\nabla\|\theta\|_{2,b}^b$. Every neuron is then balanced and earns its cost share: $T_q(\omega_i) = (h/P)\|\omega_i\|^b$ for the multilinear form $T_q(\omega) = \mathbb{E}_q[g_\omega]$. Writing t_i for T_q at neuron i with unit factors and $a = 1/(s-1)$, this gives the exact factorisation

$$\gamma(\theta) = \nu^{-\nu/2}\Big(\sum_i t_i^{-a}\Big)^{-1/a}.$$

For a single feature the gain over IF splits into a *width factor* $(\nu/m)^{s-1}$ and a *critical-value factor*. At IF optima the identity is the dual form $\Gamma_s = (\sum_{j\in S} K_j(\mu^\star)^{-a})^{-1/a}$ of the programme. Three consequences:

(i) Every certificate q independent of θ only recovers the dense value: $\inf_q \sup_\theta \mathbb{E}_q g_\theta/\|\theta\|^\nu = B_p\Gamma_1(p)$ for every b. A sharp dual must depend on θ.

(ii) If some KKT measure is uniform in x, every neuron is frequency-diagonal, and a global maximiser then has exactly $|\operatorname{spt} y^\star|$ features. But uniformity of q is equivalent to interference-freeness, so this is circular.

(iii) Local maxima violate the count at $p \geq 5$. A polished network at $p = 5$, $n = 6$, $b = 2$ (six neurons, $\nu = 7$) is a local maximiser with $\gamma = 0.265\,\gamma_{\mathrm{IF}}$. It passes a numerical second-order check: the reduced Hessian has eigenvalues in $[-0.20, -0.012]$ on the critical cone. Its feature 1 is cheap ($|\beta_1|/a_1 = 1.099$), and it carries a spurious feature 2 with $|\beta_2|/|\beta_1| = 1.3 \times 10^{-4}$, although $K^* = 1$. We recomputed the margin and amplitudes independently.

So first- and second-order conditions do not exclude cheap or spurious features. Any proof of the count must use the *global* input $\gamma(\theta^\star) \geq \nu^{1-s}B_p\Gamma_s$, which is exactly how the master inequality uses it. Through the factorisation that input becomes a contest between the width gain and the critical-value loss. These are p-dependent rates, which is the same wall seen from the dual side. The KKT measures of the $p = 3$ counterexample and of the $p = 5$ local maximiser both concentrate on the cone-edge directions and are far from uniform; they show no qualitative difference. Files: `mechanism_scripts/kkt/`.

How the four gaps mesh.

1. *One combinatorial hypothesis removes both (R) and the free part of (C).* If the doubled support $2S = \{2l : l \in S\}$ has no relation $\sum_l c_l\, 2l \equiv 0$ with $c_l \in \{0, \pm 1, \pm 2\}$ (one-dimensional dissociation), then every assignment of sets T_l satisfies (D), because a relation in $\mathbb{Z}_p^n$ is a relation in each coordinate. For the same reason no signed sum lands on a feature's own on-frequency mode. If moreover no feature is ± 3 times another, (F2) cannot occur. What is left is *paid* (off-frequency) content, at the chosen characters or at their signed sums (Lemma 27), plus the own partner (F1). For a single cheap feature this is a finite game for each (p, s). The network picks the size R of a cheap feature's sign-flipped mode, its partner b and foreign modes z_j, z_j' from the other features. It gains at most $\frac{1}{2}RK_l(\mu)$ by Theorem A and pays $K_j(\mu)\big(a_j - \sqrt{a_j^2 - |z_j|^2 - |z_j'|^2}\big)$ per riding feature. The prover picks μ, and detects $\sum_\delta \mu_\delta \big|\frac{1}{2}(a\,c_l(\delta) + b\,\overline{c_l(\delta)}) + \frac{1}{2}\sum_{j\neq l}(z_j c_j(\delta) + z_j'\overline{c_j(\delta)})\big|$. By Proposition 22, exact hiding needs $|B|+1$ coefficients, while riding supplies at most $2|S| \leq |B|$.

2. *Large p forces (R).* One-dimensional dissociated sets in $\mathbb{Z}_p$ have at most $\log_3 p \approx 0.91 \ln p$ elements, fewer than $K^* \approx 1.3$–$1.5 \ln p$. So at large p relations exist, and (O1) cannot be avoided. The relations it must survive are long. With weights ε, the Riesz coefficients have total mass $\sum_\sigma \prod(\varepsilon/2) = (1+\varepsilon)^K$, so if the signed sums of a K-set equidistribute, the weight landing on a given character is about $(1+\varepsilon)^K/p$. At $\varepsilon = 1$ this is $2^K/p$: $p^{-0.03}$ at $K = 1.4 \ln p$ and $p^{+0.04}$ at $K = 1.5 \ln p$. The plain Riesz product is thus borderline exactly at the programme's scale, and $\varepsilon < e^{\ln p/K} - 1$ (≈ 0.95 at $K = 1.5 \ln p$) restores decay at the price of extraction. This is why (O1) is a question of constants. The three-term relations of (F2) are rare by comparison: a random set of that size has about K^3/p.

3. *(O3), (O4) enter only through the constant in (P).* At $n = 2$ the surplus of detection over payment per feature and direction is $|\beta_{l,T}|\big(\varepsilon\sqrt{\kappa_l/2} - \lambda\kappa_l\big)$. With the proved $\lambda = \frac{1}{2}$ and $\varepsilon = 1$, its ratio to the payment is $\sqrt{2/\kappa_l} - 1$, which tends to 0 as $\kappa_l \to 2$; in the worst direction the surplus is $O(p^{-2})$. With the sharp single-feature constant $c_1(2,s) = 1 - 3^{1-s}$ (Theorem 34), concavity in κ gives a surplus of at least $(\varepsilon - 2c_1)\frac{\kappa_l}{2}|\beta_{l,T}|$ in every direction: 18% of the payment at $b = 2$, $\varepsilon = 1$. This surplus is the common currency: relations are absorbed by lowering ε (or by Fejér kernels), unpaid compensation by the residual of the game above, and both have to fit inside it. With linear tests a uniform surplus needs the sharp pointwise constant at $n = 2$, a single-feature statement of type (P). Quadratic tests give one from Theorem A alone: at least 39% of the detection, in class $\mathcal{C}$ under (D_3) (Corollary 30 and remark (a)).

4. *For $n \geq 4$ the factorisation must change, and the degree of the test is the exchange rate between (O1) and (O3).* Pointwise (P) is false at $n = 4$, $b = 2$ ($c_1^{\mathrm{cf}} \geq 1.127$), so the payment must be of L^2 type. Parseval (A_2) loses $\sqrt{|L'|}$. Quadratic products (Theorem 29) do not, at the price of order-three dissociation. They need $k^\sharp(n,s) < 1$, or the smaller direction-resolved Hermitian-square constant $k^\flat(n,s) < 1$ (remark (d)). Local search at $b = 2$: $k^\sharp \geq 0.514,\ 0.848,\ 1.196,\ 1.488$ for $n = 2, \dots, 5$ (proved ≤ 0.608 at $n = 2$), and $k^\flat \geq 0.423,\ 0.535,\ 0.559,\ 0.570$ for $n = 2, \dots, 5$. These are lower bounds from local search; nothing is proved for $n \geq 3$, but the Hermitian-square constant stays far below 1 where the quadratic one does not. The quadratic factor alone fails from $n = 4$ on, and cannot be rescued at large n: $k^\sharp \geq k^*$, and $k^*(n, \nu/2)$ increases numerically (0.35 at $n = 2$ to 1.04 at $n = 9$). As the degree d of the per-feature factor grows, its detection increases to $-\min_x X_l$, and the per-feature condition becomes

   (S_p) $\quad \frac{1}{2}\sum_\pm \min_x \mathrm{Re}\left(e^{\pm i\alpha}\Phi(x)\right) \leq \nu^{1-s} Q^s \cos\alpha$ for every single-feature atom system and $\alpha = \frac{\pi}{2} - \frac{\pi k}{p}$, 1

   At the one direction $\alpha = \frac{\pi}{2} - \frac{\pi}{p}$ it implies (Q1), because $\cos\alpha \, \min_x \mathrm{cm}(\Phi) = \min_\pm \min_x \mathrm{Re}(e^{\pm i\alpha}\Phi)$ is at most the average over $\pm$. So in a product architecture, (O3) and (O4) ask for (S_p) with room, certified at some finite degree. A degree-d product factorises only under order-$(d+1)$ dissociation, whose capacity is smaller: at most $\log_{d+2} p$ elements (the sums $\sum_l \varepsilon_l l$ with $0 \leq \varepsilon_l \leq d+1$ must be distinct), against $\log_3 p$ for $d = 1$. So the two kinds of open problem pull in opposite directions. (O3) and (O4) improve with the degree of the per-feature test; (O1) gets harder with it. For $n = 2, 3$ at $b = 2$ the quadratic level is enough on the (O3) side.

5. *How the claim itself could fail.* Neither relations (O1) nor compensating modes (O2) create features. A clean amplitude at q needs frequency q in every input slot and in the output, and riding modes, stray outputs and partners never have that (Lemma 26). If cheap features did occur at a maximiser through (O1) or (O2), the count could change only through a shift

of the support of the clean allocation away from every programme maximiser. A failure of (O4) is more direct: a cheap single feature at $p = 5$ with multi-frequency inputs could carry a spurious feature (paragraph "spurious features need all-slot content").

Status of the feature-count theorem

Claim. For every odd prime p, every n and $b \in [2, \nu]$, every global maximiser of the $L_{2,b}$-margin uses $|\operatorname{spt} y^\star|$ features for some maximiser $y^\star$ of $\Gamma_s(p)$. (Several maximisers with different support sizes can coexist at isolated values of s, so the claim cannot be sharpened to a single K^* there.)

Proved. (i) $p = 3$ (one frequency). (ii) Every maximiser that buys no feature below its interference-free price (Proposition 18), for all n, p, b. (iii) The dense range of Theorem 8. (iv) $n = 2$: no single feature can be bought below the interference-free price, at any prime and for arbitrary inputs (Theorem 20). Cheap features in a multi-feature maximiser force an additive relation among their doubled frequencies, or a mode landing exactly on a sign-flipped character (Theorem 19). Exact hiding from a programme vertex is impossible with the features' own outputs (Proposition 22). In class $\mathcal{C}$, few cheap features or small $\sum \kappa$ suffice (Theorem 17). (v) For each n, single-frequency single features cannot beat the interference-free value at large p (Corollary 14). (vi) Class $\mathcal{C}$, $n = 2$, $b \in [2, 3]$, every $p \geq 5$: any number of cheap features whose frequencies satisfy the order-three dissociation (D_3) and isolation (I_2) cannot beat the interference-free value, with a surplus of at least 39% in every direction (Corollary 30). For every n the same holds as soon as the atom constant $k^\flat(n, s)$ is below 1. (vii) Class $\mathcal{C}$, $n = 2$, $b \in [2, 3]$, $p \in \{5, 7\}$: every maximiser within $\mathcal{C}$ is interference-free and uses $K^*(s, p)$ features, with no condition on the cheap set (Corollary 32(c)). In the relations case (all interference on $T = \{1, 2\}$) the same holds for every $p \leq 59$, and at every prime when the cheap set carries $\sum \kappa_l < 29.7$ (Corollary 32, Theorem 33). (viii) $n = 2$, single-feature payment, sharp and p-free: P_{12} (Theorem 33) and $c_1(2, s) = 1 - 3^{1-s}$ (Theorem 34, computer-assisted). (ix) $n = 2$, every prime, any number of cheap features: near-IF rigidity (Theorem 36). In the relations case and in class $\mathcal{C}$, no network whose interference energy is below a fixed fraction of its smallest cheap atom mass beats the interference-free value. The fraction does not depend on p. It rests on quadratic payments (Proposition 35; (ii) computer-assisted). In the relations case, what is left needs $\sum_{L'} \min(c_{12}, K_q t_l)^2 \kappa_l \geq 1$: dozens of cheap features with macroscopic interference.

Numerical evidence on cheap features at $p \geq 5$ (local searches; files in `mechanism_scripts/cheap/`).

- *Single features, arbitrary inputs* (slot-symmetric, seeds: IF, IF plus a frequency-$3l$ component, random). The best efficiency is exactly the IF value at $p = 5$ for $n = 2, 3, 4, 5, 8, 9, 10, 11, 12$ and at $p = 7$ for $n = 2, 3, 4$. No width below ν is feasible, even where the shared-sign bound $\nu - \lfloor \nu/5 \rfloor$ would allow it ($n \geq 9$). This suggests that at $p \geq 5$ every positive-margin single-feature network needs ν neurons, as interference-free ones do. This includes $p = 5$, $n = 8, 9$, where $k^*(n, \nu/2) \geq 0.994, 1.042$ exceeds k_5 and Theorem 14 of the companion note gives nothing.

- *Parasitic orbits and global searches* ($n = 2$, $b = 2$). With a feature forced cheap the best values are:

  - $p = 13$: 0.90 ($K^* = 3$);
  - $p = 17$: 0.78;
  - $p = 19$: 0.97.

Given free extra neurons, the optimiser *repairs* the cheap feature back to price ratio 1.000. Global searches from IF with extra neurons stay at price ratios 0.9999–1.000 at $(7,2)$, $(11,2)$, $(7,3)$ and $(5,3)$.

- *Spurious features in sub-optimal networks.* These appear mainly at the doubled frequency $2l$ of a cheap feature, with relative amplitudes 10^{-3}–3×10^{-2}, and vanish when the feature is repaired. This is the "multiples" condition of the Fejér remark at work.

- *Falsification at large p* (relations model, $n=2$, exact oscillation over $\mathbb{Z}_p$, all allocations, cheap sizes up to the cap, optimised phases; `mechanism_scripts/falsify/`). At $b=2$ no configuration with a cheap feature reaches the IF value for any p from 31 to 1009. Let c^* be the smallest payment constant at which some configuration found would win. It is $0.58, 0.65, 0.68, 0.74, 0.75, 0.76$ at $p = 31, 61, 151, 307, 503, 1009$: rising, and 3–4× the proved $c_{12} = 0.1835$. The best realisable cheap networks fall from 0.57 to 0.46 of IF over the same range, and the exact margins over $\mathbb{Z}_p^2$ match the model.

  - *Seeds.* Relation-rich seeds (arithmetic progressions with chirp phases, Bohr sets, multiplicative cosets, Legendre-symbol phases) do worse than random or programme-based supports.
  - *Parseval.* The perfect-flatness floor falls like $0.86/\sqrt{\ln p}$, so a purely L^2 (Parseval) proof works with Theorem A's constant only up to $p \approx 7\cdot10^3$. With c_{12} it works up to $p \approx 3\cdot10^9$ (extrapolated). Beyond that, flatness arguments are needed, and they have a factor of about 4 of room.
  - *Near $b=3$.* The closest approach is there: full-F Legendre-phase pairs reach 0.88–0.91 of IF at $s = 1.05$–1.1. In that range Theorem A's constant would actually fail ($c^* < 0.289$), but the sharp $c_{12}(s) \to 0$ has $c^*/c_{12} \geq 2.9$ at every (s,p) tested. So Theorem 33 is needed there, and suffices.
  - $p=5$, $n = 8, 10, 12$. Inputs shared within groups of slots (frequency-2 content, non-symmetric noise, constant components) all return to IF. Sign-vector starts reach at most 0.13 of IF.

Open, stated as self-contained problems.

(O1) *Relations: the oscillation form.* (Corrected; the one-sided form first stated here is false.) For all allocations $a \geq 0$, interference sizes $0 \leq v \leq 3^{s-1}a$ and phases, $\min_{\delta\neq0}\big[\sum_l \kappa_l(\delta)(a_l + c\,v_l) - \frac{1}{2}\operatorname{osc}_w D_\delta\big] \leq \Gamma_s(p)\,(\sum_l a_l^{1/s})^s$, with $c = 1 - (\frac{2}{3})^{s-1}$ (P_{12}, Theorem 33). This is the relations case at $n=2$. It is proved for every $p \leq 59$ and, at every prime, for cheap sets with $\sum_{L'} \kappa_l < 29.7$ (Corollary 32, Theorem 33). It remains open for cheap sets spread over most of F at $p \geq 61$. By Proposition 37 only allocations with efficiency above $\frac{2}{3}$ matter, and for those it is a flat-polynomial bound on near-optimal supports. Moduli-only arguments provably cannot reach it under Theorem A's constant. With P_{12} (now proved) it follows from a relation-mass hypothesis (RM) for efficient allocations, verified numerically up to $p = 10007$. Mixed sets T add the factor of Theorem 17, and the multi-T product tests of Theorem 29 need order-three dissociation.

(O2) *Compensation costs.* Show that modes placed on a cheap feature's sign-flipped characters by other outputs never cost less than the bonus they protect. By Lemmas 26 and 27, every such mode is either paid quadratically through U (riding, stray outputs, dedicated neurons), or is the own partner, or needs a shared neuron and a relation $2l \equiv \pm j \pm k$ in the support. Exact

first-order hiding is possible iff the channels supply $|B|$ output frequencies (Proposition 28), and stray outputs supply them at every sparse prime. So linear gain bounds cannot close (O2), and the gain's quadratic behaviour near the IF block must be used. For one cheap feature the game is won numerically at every $p \le 41$ (ratio ≥ 1.05, ≥ 1.69 with the quadratic test). What remains: a T-resolved sharp payment inequality (single-feature, p-free), a robust uncertainty principle on $\mathbb{Z}_p$, and several cheap features at once.

(O3) *Single features for $n \ge 3$.* Rigorous bounds on $k^*(n,s)$ (circularity route) or on the joint constant $J(n,s,p)$ of Proposition 21. The pointwise route alone cannot work at $n=4$, $b=2$, where $c_1 \ge 1.13$ (certified). For the multi-feature product architecture the relevant single-feature quantity is the atom constant $k^\flat(n,s) < 1$ (Corollary 30, remark (d)), and in the limit of high-degree tests the inequality (S_p) of Section 9, item 4. A partial rigorous fact towards the width conjecture: in class $\mathcal{C}$ a single feature of width $N \le n$ has $\min_x \mathrm{cm}(\Phi) \le 2^n \sin(\frac{\pi}{2p}) \sum_i |\beta_i|$. (Write $\Phi = \sum_i \overline{\beta_i} \prod_t (1 + \overline{\xi_{it}} z_t)$ with $z_t \in \mu_p$, assign each atom a slot of its own, and choose z_t there as the root of unity nearest $-\xi_{it}$. That factor is at most $2\sin(\pi/2p)$ and every other factor is at most 2.) So width below ν loses to the IF value once $2^n \sin(\pi/2p) < \nu^{1-s}$, i.e. for $p \gtrsim \pi 2^{n-1} \nu^{s-1}$. This says nothing at $p=5$.

(O4) *$p=5$, large n.* $k^*(n,\nu/2) > k_5$ from $n=8$ on, and $J(n,\nu/2,5)$ rises $(0.57, 0.71, 0.81, 0.82$ for $n = 3$–$6)$. The strongest known mechanism (shared-sign) decays at $p=5$ ($\times 0.93$ per unit n up to $n=18$), and direct searches at $n=8$ find nothing above interference-free. If a mechanism with multi-frequency inputs did win at some $(5,n)$, it could create a spurious feature and break the claim. So (O4) decides whether the claim is true for all n.

The route here proves the claim by excluding cheap features, which at $p \ge 5$ amounts to the paper's Conjecture 20 there. A proof that tolerates cheap features would have to establish decoupling (equal effective prices for all features) instead. Either way (O1)–(O4), or statements of the same strength, are needed.

Statement	Status
$\gamma_b^* = \nu^{1-s} B_p \Gamma_s(p)$ for all p, n, b (Conjecture 20)	**false**: $p = 3$, $b = 2$, $n \geq 11$; excess 2.21× certified at $n = 18$, growing
$\nu^{1-s} B_p \Gamma_s \leq \pi_{n,b}(p) B_p \Gamma_s \leq \gamma_b^* \leq B_p \Gamma_s$	proved (companion note)
$\gamma_b^* = \pi_{n,b}(3)\, B_3 \Gamma_s(3)$ at $p = 3$	proved (one frequency)
width $\geq \nu - \lfloor \nu/p \rfloor$ for shared-sign networks	proved (Theorem 3)
single-frequency networks cannot beat IF when $k_p \geq k^*(n, s)$	proved (companion note); k^* numerical
$\gamma_b^* = \pi_{n,b}(p) B_p \Gamma_s(p)$ (Conjecture 20′, decoupling)	open; genuine obstacle exhibited (Section 6)
$\pi_{n,b}(p) = \nu^{1-s}$ for $p \geq 5$	open; every mechanism tested loses, by a factor growing in n
feature count of every global optimiser $= K^*(s, p)$, $p = 3$	proved (Prop. 6(a))
feature count $= K^*(s, p)$, all p, dense range of Thm. 8	proved (Thm. 8; extends $[b_0, \nu]$ in all cases computed)
feature count $= K^*(s, p)$, all p, given decoupling	proved (Prop. 7); price irrelevant
feature count $= \Theta(\log p) = \Theta(K^*)$, all p, unconditional	proved (Thm. 9 with paper's Cor. 12)
single-frequency networks, $n = 2$, all $p \geq 5$, $b \in [2, 3]$: no gain over IF	proved (Cor. 10)
$\lvert\beta_j\rvert \leq \nu^{1-s} Q_j^s + K_{\nu,s} \ell_j$ (Lemma 7 with interference), every n	proved (Lemma 13)
single-frequency networks, every n, b, all $p \geq p_0(n, b)$: no gain over IF	proved (Cor. 14)
count $= K^*$ in class $\mathcal{C}$, $n = 2$, all p, $b \in [2, 3]$, ≤ 2 features below IF price	proved (Thm. 17)
count $= K^*$ in class $\mathcal{C}$, $n = 2$, $p \in \{5, 7\}$, $b \in [2, 3]$, no condition	proved (Cor. 32)
relations case ($T = \{1, 2\}$), $n = 2$: every $p \leq 59$; any p if $\sum_{L'} \kappa < 29.7$	proved (Cor. 32, Thm. 33)
sharp $n = 2$ payments: P_{12}, and $c_1(2, s) = 1 - 3^{1-s}$	proved (Thm. 33; Thm. 34, computer-assisted)
count $= K^*$ for every maximiser with no feature bought below IF price, all n, p, b	proved (Prop. 18)
count $= K^*$ in class $\mathcal{C}$, $n = 2$, all $p \geq 5$, $b \in [2, 3]$, any number of cheap features under (D_3), (I_2)	proved (Cor. 30)
$n = 2$, any network: cheap features force additive relations or spectral coincidences	proved (Thm. 19)
$\pi_{2,b}(p) = 3^{1-s}$ for every prime p (single features, arbitrary inputs): (Q1) at $n = 2$	proved (Thm. 20)
feature count $= K^*(s, p)$, $p \geq 5$, $b < b_0$	open; reduced to relations among cheap features and off-frequency compensation (Sec. 8)

Answer to "prove that the unrestricted optimum matches the prediction". In the interference-free form this cannot be done: the statement is false at $p = 3$, and increasingly so

as n grows. In the price–programme form it holds at $p = 3$ by definition, and there the object to compute is $\pi_{n,b}(3)$. We have shown that, in the class that contains every counterexample, this reduces to choosing $\nu - \lfloor \nu/3 \rfloor$ real slopes, with the weights given by an LP. For $p \geq 5$ the full statement would follow from two claims:

(Q1) $\pi_{n,b}(p) = \nu^{1-s}$, i.e. a single feature cannot be bought below the IF price. Theorem 3 limits the width saving of the $p = 3$ mechanism to $\lfloor \nu/p \rfloor$ neurons, and to none for $p > \nu$. The circularity theorem of the companion note handles single-frequency inputs when $k_p \geq k^*$. A proof needs a cost-weighted (robust) form of the width theorem for arbitrary sign patterns.

(Q2) Decoupling. It has to control compensation by interference-only fields, which demonstrably raises a feature's margin.

Both hold in all our experiments at $p = 5, 7$, by margins of $4\times$ or more that grow with n.

Files. `certify_1d.py` (rigorous certificate for the 1-D family, any p), `oned_certified/` (parameters of the networks in Table 1), `mechanism_scripts/` (the field model, the LP, the slope search and continuation, the joint two-frequency LP, the release test).